BlinkTwin: An Open, In-Browser Model of Subjective Sleepiness That Anchors Webcam and Reaction-Time Signals to Self-Ratings

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Abstract*—*Objective: **To estimate subjective sleepiness on an ordinary laptop by combining a biomathematical model, occasional self-ratings and passive signals, and to test the approach on public data.** ***Methods:*** **BlinkTwin runs in a web browser; video never leaves the page. A Kalman filter tracks the person's offset from the Three-Process Model of alertness, updated by Karolinska Sleepiness Scale (KSS) check-ins. Reaction time, blink duration and PERCLOS enter through forward models fitted within person on DROZY (14 adults), each with a personal anchor state. We replayed the model on DROZY leaving one subject out (with landmark eye features), ran the website's camera pipeline on DROZY's near-infrared video against the electro-oculogram (EOG), analysed OpenNeuro ds004902, and ran a simulation with a consider-covariance comparator and misspecified scenarios.** ***Results:*** **On DROZY, errors were lower with each source (mean absolute error 1.73 KSS points for the prior, 1.38 with a check-in and passive signals), but intervals overlapped and a protocol-aware baseline gave 1.39. The camera pipeline's blink rate correlated with EOG ($\rho = 0.71$) but undercounted blinks, blink duration tracked KSS within person except in the 30-frame-per-second subset, and PERCLOS failed. On ds004902, a group-mean offset predicted better than a personal one. In simulation, estimating the anchors lowered error by 0.04 points against the consider-covariance filter, which was no better than check-ins alone; in a simulated 10-day study, passive signals improved the prediction of observed ratings by 0.01 points, and by up to 0.05 with a mean-reverting offset.** ***Conclusion:*** **Check-ins carried most of the information; passive signals added little, and only when their personal biases were estimated.** ***Significance:*** **An open, on-device implementation, a transparent public-data evaluation, and a planned validation protocol.**

***Index Terms*—Kalman filter, Karolinska Sleepiness Scale, PERCLOS, psychomotor vigilance test, sleepiness, state-space model, Three-Process Model, webcam.**

I. Introduction

SLEEP loss impairs performance in ways that matter for safety. In a classic comparison, 17 h of sustained wakefulness reduced performance on a tracking task to the level seen at a blood alcohol concentration of 0.05%, and 24 h to the level seen at 0.10% [1]. People are also poor judges of the impairment that builds up over days of short sleep: under chronic sleep restriction, lapses on the psychomotor vigilance test (PVT) keep increasing while sleepiness ratings level off [2], and individual differences in subjective and objective vulnerability to sleep loss are stable but unrelated [3].

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Code, analysis scripts, result files and the scripts that produce every figure and table: https://github.com/r11up/BlinkTwin. The version described here is tagged `v1.0-arxiv`.

Three kinds of tool address parts of this problem. *Biomathematical fatigue models* predict sleepiness or performance from the sleep–wake history and the time of day [4]–[9]. They need no sensor, and some of them are individualised from repeated performance tests [10]–[14]. *Ocular measures* such as PERCLOS and blink duration track drowsiness [15]–[18], and blinks can be detected in ordinary video [19], but most systems map them to drowsiness with one rule for everyone, although the link between eyelid signs and sleepiness differs between people [20]. *Validated questionnaires*, such as the Karolinska Sleepiness Scale (KSS) [21], capture how sleepy a person feels and track performance and EEG changes [22], [23], but they can be asked only a few times a day.

This paper asks whether an ordinary laptop can combine all three: keep a continuously updated estimate of how sleepy a person feels, state its uncertainty, and project it forward, with video kept on the device. We call the result a personal digital twin in the sense of a person-specific model that is updated from that person's data and can be queried for forecasts and what-if scenarios [24], [25]. The model itself is a small state-space model, and we use that term in the technical sections.

The difficulty lies in combining sources that disagree. A webcam or reaction-time signal can read a particular person consistently high or low. If such a signal is averaged with the person's ratings, it pulls the estimate away from what the person reports. BlinkTwin therefore gives each passive signal a forward measurement model with its own personal *anchor*: a slowly drifting bias state, in the signal's own units, that absorbs where this person's signal sits relative to the population. Check-ins set the level of the estimate, and passive signals contribute mainly through their changes. Bias states in a Kalman filter are a standard technique [26]; the contribution here is to apply them to self-rated sleepiness and to test the result.

Because the KSS is the reference, BlinkTwin estimates *subjective* sleepiness. Given the dissociation described above, the application also shows the latest reaction test read on the population scale, and it flags a possible gap only when the person's own reaction times slow over a day or a week while their ratings do not.

The short answer from public data and simulation is that check-ins and the biomathematical prior do most of the work. Passive signals add a little, and only when the model estimates each person's bias in each signal; a filter that merely allows for those biases gains nothing. The website's camera pipeline measures blinks on laboratory video but not PERCLOS, and

the passive signals' benefit has yet to be shown in real use. We report these limits alongside the method because they decide what the planned study has to test.

Contributions:

1) An open, on-device implementation that joins the published Three-Process Model of alertness [6], [27], [28] with KSS check-ins and forward models of a 3-min PVT (PVT-B), blink duration and PERCLOS in one Kalman filter, with a personal anchor for each passive signal (Sections III and IV).
2) An evaluation on public data: a leave-one-subject-out replay of the model on DROZY [29]; a test of the website's camera pipeline on DROZY's near-infrared video against the electro-oculogram (EOG), which exposed a failure of PERCLOS that we then guarded against; and tests of the offset premise and of the prior on OpenNeuro ds004902 [30] (Section V).
3) A simulation study, reported following Morris et al. [31], with a consider-covariance comparator [32], a known-parameter reference, scenarios that break the model's assumptions, and a sensitivity analysis of the filter settings (Section VI).
4) A planned protocol for a prospective validation study, to be registered before data collection (Section VII).

BlinkTwin is a self-monitoring and research tool for desk work. It is not a medical device, it must not be used to decide whether someone is fit to drive or operate machinery, and the absence of an alert does not indicate alertness.

II. Related Work

A. Subjective scales and vigilance tests

The KSS asks for a rating of sleepiness over the last few minutes on a 9-point scale from "extremely alert" to "very sleepy, great effort to keep awake, fighting sleep" [21]. Ratings rise with time awake and at night and go with slower reactions and EEG signs of sleepiness [22], [23]; the original version labelled every other step, a later version labels all nine, and the two have been compared directly [33]. The Stanford Sleepiness Scale (SSS) is an older 7-point alternative [34]. The PVT measures simple reaction time to a stimulus at random intervals [35]; lapses and response speed are its most sensitive outcomes [36]. The 3-min PVT-B uses inter-stimulus intervals of 1–4 s and a 355-ms lapse threshold and detects total and partial sleep deprivation [37], and 3-min PVTs also work on phones and tablets [38]. Web browsers add display and input latencies of tens of milliseconds that vary between devices [39], [40].

B. Ocular measures and video

PERCLOS, the proportion of time the eyelids are at least 80% closed, came from a programme of driving-simulator studies [15] and was the ocular measure most closely related to PVT lapses in a later comparison [16], [41]. Blink duration and the eyelid closing and reopening times increase with sleepiness [17], [18], and infrared reflectance oculography detects drowsiness-related impairment against PVT and driving outcomes [42], [43]. In video, the eye aspect ratio from facial landmarks detects blinks [19]; MediaPipe provides real-time face meshes in a browser [44], [45] and eye-closure blendshape coefficients [46]. Public datasets with real, not acted, drowsiness include DROZY [29] and UTA-RLDD [47]; the DROZY group later trained face-video models on a larger near-infrared dataset to predict drowsiness defined from PVT reaction times [48]. Such "microsleep" detectors flag long eye closures; true microsleeps are defined by EEG [49].

C. Biomathematical models and individualisation

The two-process model combines a homeostatic sleep pressure with a circadian rhythm [4]. The Three-Process Model of alertness adds sleep inertia and an ultradian component and maps alertness onto the KSS [5], [6], [50]; it was later given a breakpoint in sleep recovery to account for partial sleep deprivation [27] and was validated and extended in airline operations, where the between-person standard deviation (SD) of KSS around the model was 0.84 [28]. Models used in transport and defence scheduling were compared in [8], [9]. Individualisation from repeated PVTs uses nonlinear mixed-effects or Bayesian methods [10]–[12] and runs in real time in the Unified Model of Performance and the 2B-Alert app [13], [14], [51]. Sandberg et al. combined a Three-Process-Model-based predictor with driving-behaviour indicators and found the combination better than either alone [52], and Ahlstrom et al. added an eye-movement test to the same predictor to forecast whether drivers would reach KSS 8 within 2 h [53].

BlinkTwin differs from these models in three ways. It targets the person's own rating of sleepiness rather than PVT performance; it adds passive webcam signals between tests; and it learns a separate anchor for each passive signal so that the person's ratings stay the reference. It does not yet individualise the prior's parameters, which the cited methods do. State-space models of intensive self-report data are also established in psychology [54]; BlinkTwin applies the idea to one rating scale with a physiological prior.

III. System

BlinkTwin is a single-page web application written in TypeScript (Fig. 1). It needs a laptop with a webcam and a current browser, installs nothing, and runs all computation in the page. The main screen (Fig. 2) shows the current estimates of sleepiness and workload with their 90% ranges, the forecast, how each source has moved the estimate, and the day so far with the check-ins, reaction tests and webcam periods. A demonstration mode loads a simulated day without touching the person's own data, and a study mode hides every estimate (Section VII).

A. Webcam sensing

Frames are passed, at up to 24 per second, to the MediaPipe Face Landmarker [44], [45] in video mode with blendshapes and the facial transformation matrix enabled. The raw eye-closure score c_t of a frame is the mean of the `eyeBlinkLeft` and `eyeBlinkRight` blendshapes [46] (0 open, 1 closed), or missing when no face is found. Because the

TABLE I
PUBLISHED SYSTEMS FOR RELATED PROBLEMS AND THEIR REPORTED RESULTS, WITH BLINKTWIN'S RESULTS ON PUBLIC DATA

Study	Data (n)	Outcome	Inputs	Validation	Reported result
Sleepiness on the KSS scale					
Ingre et al. [28]	Airline crews, field	KSS	Three-Process Model from sleep	In-sample mixed model	Residual SD 1.46 with model-generated sleep
Knock et al. [55]	21 ICU nurses, field	KSS	Arousal model, shift schedule, sleep, light	Prospective	56–60% within ±1 point (day shifts), 48–52% (night)
Sandberg et al. [52]	12 drivers, simulator	Sleepy vs alert (KSS)	Driving indicators + sleep/wake model	Held-out data	Best single indicator 0.72 (mean of sensitivity and specificity)
Alertness on the PVT					
Reifman et al. [14]	21 adults, 62 h awake	PVT mean RT	In-app PVT + personalised model	Prospective	After 12 PVTs, accuracy comparable to the peak with all PVTs
Subramaniyan et al. [56]	26 adults, 62 h awake	PVT mean RT	Face video + personalised model	Cross-validated	Error 36 ms (SE 5) vs 30 ms rested within-person variability
Drowsiness from face video or eyelid sensors					
Maior et al. [57]	DROZY	Drowsy vs awake	Eye aspect ratio (near-infrared)	Inter-subject	94.44% accuracy
Massoz et al. [48]	29 adults (Liège)	Binary, from PVT RT	Near-infrared face video	Leave one subject out	70%, 85%, 89%, 94% (5, 15, 30, 60-s windows)
Ghoddoosian et al. [47]	UTA-RLDD, 60	3 classes, self-labelled	Phone and webcam RGB video	Subject folds	65.2% (human judges 57.8%)
Wilkinson et al. [42]	Restricted sleep	$\geq$3 or $\geq$5 lapses min^{-1}	Infrared reflectance oculography	Group	Sensitivity 77%, 100%; specificity 85%, 83%
BlinkTwin (this work)					
Replay of released model	DROZY, 14	KSS	Prior + check-in + PVT + webcam	Leave one subject out	MAE 1.38 (prior alone 1.73); Section V-C
Prior alone	ds004902, 28	KSS	Three-Process Model	Out of sample	MAE 1.69, 36% within ±1; with refitted mapping [28] 1.46, 45%
Webcam pipeline	DROZY video, 36 tests	Blink rate	MediaPipe blendshapes	vs EOG	Spearman $\rho = 0.71$

open-eye level of this score differs between people, cameras and lighting, the *blink meter* normalises it against the person's own recent baseline b_t, the 20th percentile of c over the previous 30 s:

$$\hat{c}_t = \min\left(1, \max\left(0, \frac{c_t - b_t}{1 - b_t}\right)\right). \quad (1)$$

A closure starts when $\hat{c}_t \geq 0.5$ and ends when $\hat{c}_t \leq 0.3$. Closures of 50–1000 ms are blinks; longer ones are long closures. Once a closure has lasted 2 s the baseline is frozen at its value before the closure until the eyes reopen (released after 60 s), because otherwise a closure longer than about 24 s would raise the baseline to the closed level and end itself. Every 60 s the meter reports blink rate, mean blink duration, the number of long closures, PERCLOS (the fraction of face frames with $\hat{c}_t \geq 0.8$), face coverage, the frame rate, and the median peak closure of the minute's blinks. These one-minute numbers are the only camera data stored.

PERCLOS guard: If the camera's blinks do not reach full closure on the normalised scale, PERCLOS cannot register closed eyes. This happened in DROZY's near-infrared video (Section V-B). The model therefore ignores PERCLOS in any minute whose median blink peak is below 0.85, while still using blink duration.

Head pose and gaze: Pitch, yaw and roll come from the transformation matrix and are expressed relative to the median of the last 20 s. A nod is a pitch change of at least 12° that returns within 2.5 s. Gaze comes from the `eyeLook` blendshapes; the eyes count as looking down when the mean `eyeLookDown` score exceeds 0.45. Nods are stored but not used by the model.

B. Alerts

A rule-based monitor runs on every frame, independently of the model. It raises a *long eye-closure alert* (red, with sound) when the eyes stay closed for at least 1.0 s, unless the person was looking down in the 500 ms before the closure began, and a *head-drop alert* when the head has pitched forward by at least 12° for 1.0 s while the eyes are at least half closed (the sign convention of forward pitch has not yet been verified on devices). If the face is lost while an alert is active, or after the eyes have been closed for 1 s, the alert stays on. An alert clears after the face has been seen with the eyes open for 1.5 s. Amber warnings are raised when, over the last minute, PERCLOS is at least 15%, there are two or more closures longer than 1 s, mean blink duration is at least 300 ms with at least five blinks, or the model's KSS estimate reaches 7; warnings repeat at most every 5 min. These thresholds are ad hoc defaults. The rules detect long eye closures and head drops, not EEG-defined microsleeps [49], and they have not been validated.

C. Reaction test

The reaction test follows the PVT-B [37]: 3 min, inter-stimulus intervals uniform on 1–4 s, a millisecond counter as the stimulus, and a response with the space bar, a click or a tap. The stimulus time is taken when the counter is made visible,

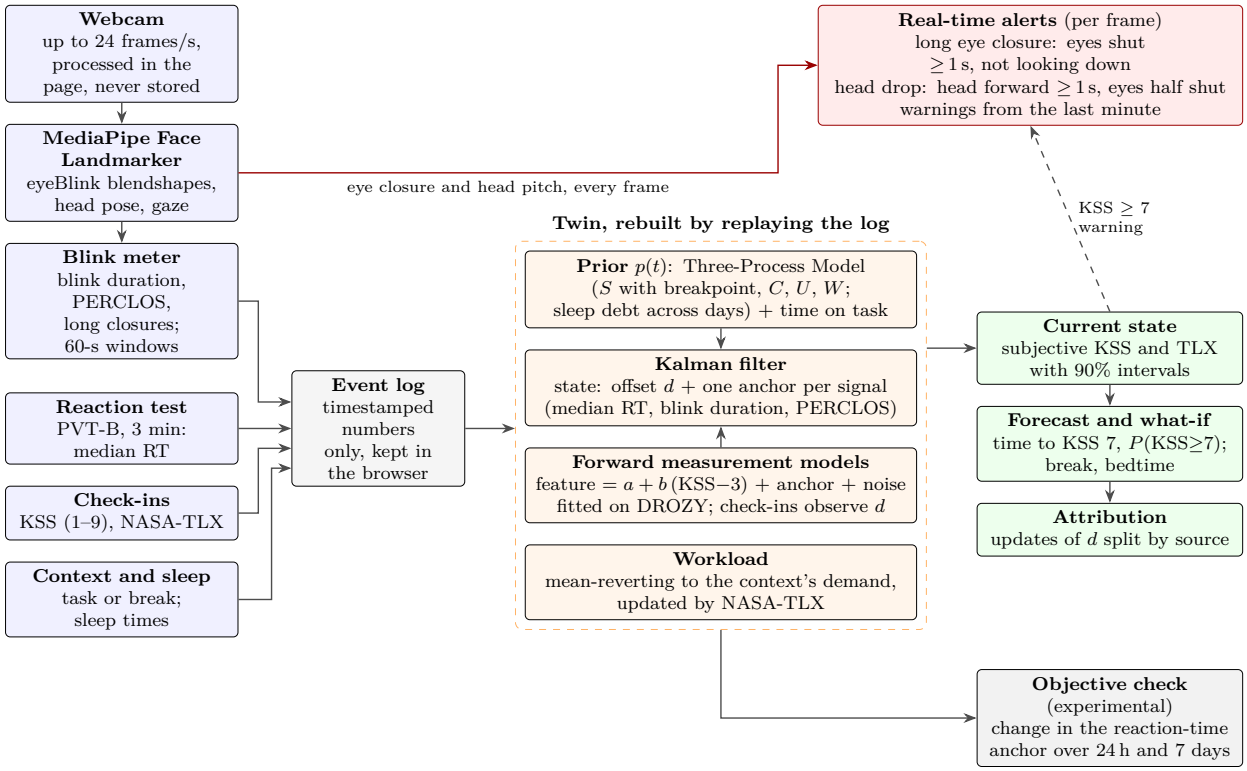


Fig. 1. BlinkTwin architecture. Webcam frames are processed in the page and discarded. Per frame, eye closure and head pose drive the real-time alerts. Per minute, a numeric summary of the eyes is stored as an event, together with reaction tests, check-ins and context entries. The model's state at any time is rebuilt by replaying these events through one Kalman filter.

and the response time from the input event's time stamp, so each reaction time includes the delay to the next screen refresh and the device's input latency. Responses with no stimulus on screen or faster than 100 ms are false starts; reaction times above 355 ms are lapses. The test reports valid responses, the median reaction time, mean response speed ($1000/\mathrm{RT}$), lapses and false starts. A 1-min version is offered for convenience; its result is shown but not used by the model.

D. Check-ins

A check-in asks for the KSS rating of the last 5 min, with all nine steps labelled [21], [33], and optionally the six NASA-TLX subscales for the work block just finished, on 0–100 sliders [58]; the raw TLX (the unweighted mean) is stored only if all six sliders were set. The person rates first and takes the reaction test afterwards, and the model's current estimate is hidden while they rate, so that neither the test feedback nor the estimate can anchor the rating.

E. Context, sleep and the event log

The person can mark what they are doing (focused work, reading, a meeting, another task or a break) and enter when they woke and how long they slept. Without an entry the application assumes waking at 07:00 after 7.5 h and says so; nights that were not entered are filled with habitual sleep and flagged, and a night without sleep can be entered explicitly. Model-driven warnings are suppressed while the estimate rests on an assumed night. The research analyses do not fill nights, so that genuine sleep deprivation is modelled as such. Every input is stored as a time-stamped event. The model has no hidden state: its estimate at any time is rebuilt by replaying the events up to that time, so every number shown is reproducible from the log, and the website, the 130 unit tests and the simulation run the same code.

F. Privacy and performance

Video frames exist only in memory for one model call. No image, landmark or blendshape trace is stored. Events are kept in the browser's local storage and can be exported (CSV or JSON) or deleted. An automated check records every network request while the camera runs, a check-in and a reaction test take place. It found that the face-tracking library sent a small usage-statistics request (timing data, no images) to a logging service every 60 s. A content security policy now limits the page's connections to its own site and the model host; with it, the check recorded no request of any kind after the page had loaded. On a 4-core cloud server without a GPU, the camera pipeline ran at 20 frames per second (mean 26 ms per face-model call) and a 10-day event log replayed in 84 ms; a laptop with a GPU should be faster.

IV. Model

The model tracks subjective sleepiness $s(t)$ on the KSS scale (1–9). It also tracks perceived workload on the raw

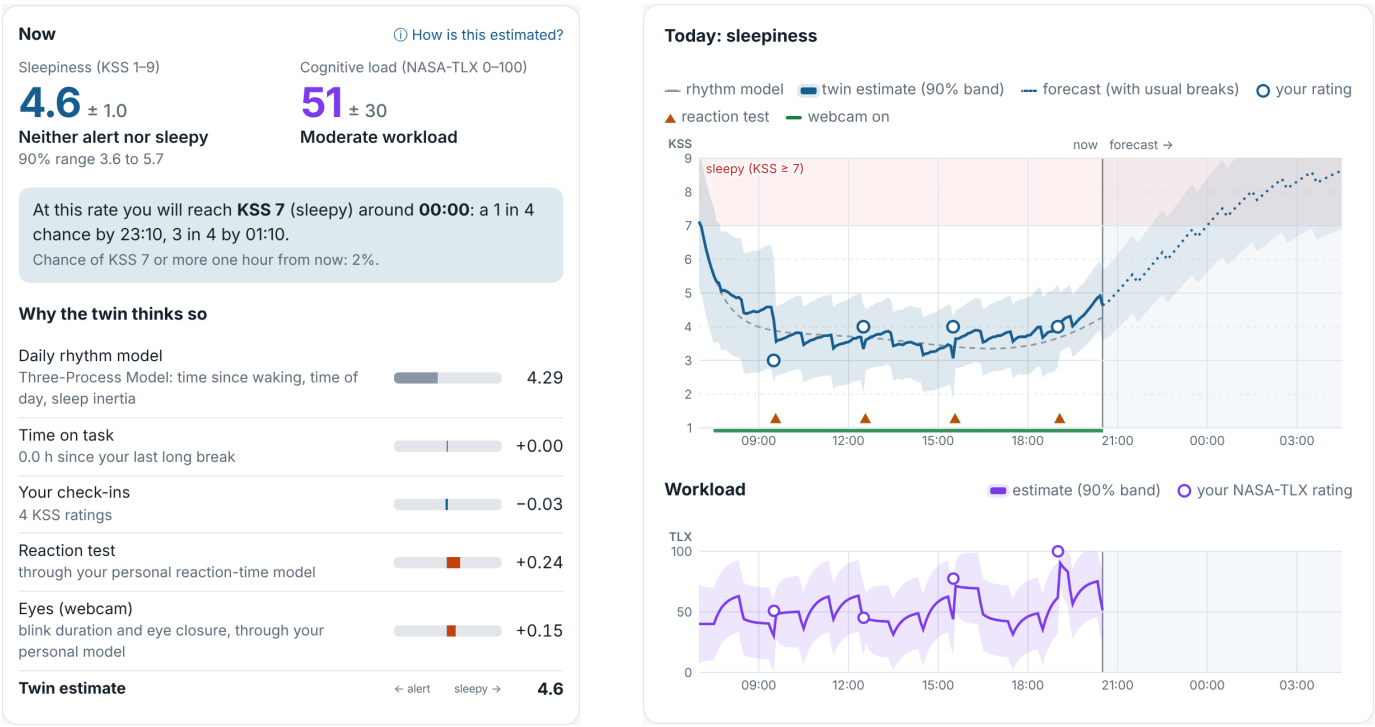


Fig. 2. The main screen on the built-in demonstration day, a simulated person (screenshots at twice the screen resolution). Left: the Now card with the sleepiness and workload estimates and their 90% ranges, the forecast of reaching KSS 7, and the estimate split into the rhythm model, time on task and the updates made by each source (Section IV); the buttons, the check-in prompt and the reaction-test panel are left out to save space. Right: the day's sleepiness timeline with the rhythm-model prior (dashed), the estimate and its 90% band, the KSS check-ins (circles), the reaction tests (triangles), the time with the webcam on (green bar) and the forecast with the person's usual breaks, and below it the workload estimate with the NASA-TLX ratings.

NASA-TLX scale; that part is a simple, unevaluated feature and is described in Appendix A. All parameters are listed in Table II.

A. Prior: the Three-Process Model

For a person who woke at time t_w, let $\tau = t - t_w$ in hours and $h(t)$ the local clock hour. Alertness is the sum of a homeostatic, a circadian, an ultradian and a sleep-inertia component [6], [28]:

$$S(t) = L + (S_w - L)\, e^{-\delta_w \tau}, \tag{2}$$

$$C(t) = M \cos\big(2\pi(h(t) - \phi)/24\big), \tag{3}$$

$$U(t) = U_m - U_a \cos\big(2\pi(h(t) - \phi_U)/12\big), \tag{4}$$

$$W(t) = W_0\, e^{-\delta_i \tau}, \tag{5}$$

$$A(t) = S(t) + C(t) + U(t) + W(t). \tag{6}$$

During sleep, S recovers linearly at rate $\delta_s(H - B)$ while it is below the breakpoint $B = 12.2$ and exponentially towards H above it [27], [28],

$$\frac{dS}{dt} = \begin{cases} \delta_s(H - B), & S < B, \\ \delta_s(H - S), & S \geq B. \end{cases} \tag{7}$$

The level at waking, S_w, is computed by running these equations over the recorded sleep episodes, preceded by ten habitual nights (7.5 h), so that a run of short nights builds up sleep debt. Alertness is mapped to the KSS and clipped,

$$p(t) = \min\big(9, \max(1,\ a_K - b_K\, A(t))\big). \tag{8}$$

For a person who wakes at 07:00 after 8 h of sleep, the prior is KSS 7.0 at waking (sleep inertia), 3.7 at 09:00, 3.1 at 16:00, 5.7 at 23:00 and 7.7 at 03:00 if they stay awake. Five consecutive 4-h nights lower S_w from 11.2 (one such night) to 8.9.

To the prior the model adds a time-on-task term $\gamma T(t)$, where T is the time in hours since the start of the current task episode; it is frozen during a break and reset once the break has lasted 10 min. The value $\gamma = 0.4$ KSS h^{-1} is an assumption; time on task raises sleepiness in driving studies [59], but the Three-Process Model was built for average sedentary workloads [6], so the term may partly double count. Its sensitivity is reported in Section VI.

B. State and dynamics

The estimate is $s(t) = \mathrm{clip}\big(p(t) + \gamma T(t) + d(t)\big)$, where d is the person's offset from the prior. The filter state is

$$x = [\, d,\ \alpha_{\mathrm{RT}},\ \alpha_{\mathrm{BD}},\ \alpha_{\mathrm{PC}}\,]^\top, \tag{9}$$

where the anchors α are how far this person's median reaction time (ms), blink duration (ms) and PERCLOS (%) sit from the population at the same sleepiness. All components are random walks with $x_0 = 0$ and $P_0 = \mathrm{diag}(\sigma_d^2, \sigma_{\alpha,j}^2)$; over Δ hours $P \leftarrow P + \mathrm{diag}(q_d, q_{\alpha,j})\Delta$. The offset drifts quickly ($\sqrt{q_d} = 0.3$ KSS h$^{-1/2}$) and the anchors slowly. Under a random walk the offset's mean is carried forward unchanged while its variance grows without limit; the code also offers a mean-reverting (Ornstein–Uhlenbeck) offset with time constant τ_d

and stationary SD σ_d, evaluated as a sensitivity analysis in Section VI.

C. Observations

A KSS check-in y observes the offset directly,

$$y - p(t) - \gamma T(t) = d + \varepsilon_K, \qquad \varepsilon_K \sim \mathcal{N}(0, \sigma_K^2). \tag{10}$$

Each passive feature f_j (median reaction time of a test, or one minute's mean blink duration or PERCLOS) has a forward measurement model,

$$f_j = a_j + b_j \big(s - 3\big) + \alpha_j + e_j, \tag{11}$$

where a_j is the population mean at KSS 3 and b_j the within-person slope. Substituting $s = p + \gamma T + d$ gives a linear observation of the state with $h = [b_j, 0, \ldots, 1, \ldots, 0]$ (1 in the position of α_j) and offset $a_j + b_j(p + \gamma T - 3)$. The noise variance is

$$R_j = \kappa_j \, \sigma_{e,j}^2 + \big(\sigma_{b,j} \, (\hat{s} - 3)\big)^2, \tag{12}$$

where $\kappa_j = (1 + \rho_j)/(1 - \rho_j)$ allows, approximately, for the lag-1 autocorrelation ρ_j of successive minutes (an explicit model of correlated noise [60] is an alternative) and the second term for between-person spread in the slope, evaluated at the current estimate $\hat{s}$. A reaction test is used only if it lasted the full 3 min. A webcam minute is used only if face coverage is at least 0.6; blink duration only if the minute has at least three blinks; and PERCLOS only if the PERCLOS guard (Section III) allows it.

Fitting on DROZY: The forward models were fitted on DROZY [29] (Section V-A) with subject fixed effects, so that the slope b_j is a within-person slope and differences in people's baselines do not leak into it. The anchor's prior SD $\sigma_{\alpha,j}$ is the between-person SD of the baselines after removing their estimation noise. For the webcam features, the per-minute residual variance was split into a within-session part, used as $\sigma_{e,j}^2$, and a between-session part, modelled as drift of the anchor, at a rate that lets the anchor wander by the between-session residual SD over about 8 h. The slope spread $\sigma_{b,j}$ is the raw SD of per-person slopes from the 8 subjects with three usable sessions, a conservative upper bound. No parameter was tuned on simulated people.

Update: Each observation is absorbed with the standard Kalman update [61],

$$k = \frac{Ph}{h^\top P h + R}, \quad x \leftarrow x + k\,(z - h^\top x), \quad P \leftarrow P - k\,h^\top P. \tag{13}$$

D. What the anchors do

A check-in pins d. A passive reading that then disagrees with the pinned level is explained mainly by its own anchor, because the anchor's prior variance is large compared with the remaining variance of d. Later readings from that signal move d only to the extent that they change relative to the learned anchor. Check-ins therefore set the level and passive signals track change. Before any check-in, the anchor and the offset share each passive update according to their prior variances, so passive signals still move the estimate, but cautiously. Two comparators use the passive signals without learning anchors. The *no-anchor* variant fixes all $\alpha_j = 0$ and adds $\sigma_{\alpha,j}^2$ to R_j as if the person's bias were independent noise on every reading, which ignores its persistence and so fails by construction; it is kept only to show that failure. The *consider-covariance* variant [32] keeps the anchors' covariance in P, so that every gain accounts for a persistent bias, but never updates the anchors' means (their gain rows are zero) and updates P with the Joseph form. It is the fair test of whether estimating the anchors, rather than only allowing for them, adds anything.

This design makes the person's ratings the reference. If a person's reaction times slow while their ratings stay flat, the model attributes the slowing to the anchor, which is the right behaviour for an estimate of how sleepy the person feels but hides objective decline [2], [3]. The application therefore also shows an *objective check*: the latest median reaction time read through the population forward model alone, $\hat{s}_{\mathrm{obj}} = 3 + (f_{\mathrm{RT}} - a_{\mathrm{RT}})/b_{\mathrm{RT}}$, labelled as not adjusted to the person. Because people's baselines differ (SD 48.7 ms, about 2.6 KSS-equivalents) and device latency adds to the level, a single reading on the population scale cannot show that a person is more impaired than they feel. The application therefore flags a possible dissociation only from within-person change, a growing reaction-time anchor over 24 h or 7 days, and presents the check as experimental and unvalidated.

E. Forecast, what-if and attribution

To forecast, the model holds the offset at its current mean and lets its variance grow, $\mathrm{sd}(t + \Delta) = (P_{dd} + q_d \Delta)^{1/2}$; the probability that latent sleepiness is at least 7 ("sleepy, but no effort to keep awake") is $1 - \Phi\big((7 - \hat{s})/\mathrm{sd}\big)$, and the predicted time of reaching KSS 7 is the first future time with $\hat{s} \geq 7$. Because that time is uncertain, the application states it together with the first times at which this probability reaches 0.25 and 0.75 (Section VI reports how well this window is calibrated). The default projection assumes the person's usual rhythm of 50-min work blocks and 10-min breaks. What-if scenarios change the inputs: a 15-min break now, then the usual rhythm; a bedtime, which replaces the sleep episode used by the prior after waking; or working on without breaks, which lets T grow. The interface also splits the offset into the sum of the updates each source has made to it (check-ins, reaction tests, webcam). This split is an accounting identity of the linear updates and depends on their order, so it is shown as a description of what the filter did, not as a causal attribution.

V. Evaluation on Real Data

All analyses use public data. DROZY results are reported as aggregates only, as its licence requires; the OpenNeuro participant table is open data. Code and result files are in the repository (Section E).

A. DROZY

Data: DROZY [29] contains 14 adults who each performed three 10-min PVTs while staying awake from the

TABLE II
MODEL PARAMETERS IN THE RELEASED CODE

Symbol	Value	Meaning and source
Three-Process Model [6], [27], [28]		
L, H	2.4, 14.3	Asymptotes of S
δ_w, δ_s	0.0353, 0.381 h^{-1}	Fall of S awake; recovery asleep
B	12.2	Breakpoint; linear recovery below
M, ϕ	2.5; 16.8 h	Circadian amplitude, acrophase
U_a, U_m, ϕ_U	0.5, -0.5; 13.8 h	Ultradian amplitude, mesor, dip
W_0, δ_i	-5.72; 1.51 h^{-1}	Sleep inertia and its decay
a_K, b_K	10.6, 0.6	Alertness to KSS
Offset and check-ins (assumed)		
γ	0.4 KSS h^{-1}	Time on task; reset after a 10-min break
σ_d, $\sqrt{q_d}$	1.2; 0.3 $\text{h}^{-1/2}$	Prior SD and drift of the offset
σ_K	0.75	Noise of one KSS rating
Forward models (DROZY, within person)		
Reaction test	a 339.2 ms, b 18.85 ms, σ_e 36.4	PVT median RT; anchor SD 48.7, drift 2 ms $\text{h}^{-1/2}$, slope SD 7.9
Blink duration	a 189.7 ms, b 11.1 ms, σ_e 24.6, κ 1.04	One minute, $\geq$3 blinks; anchor SD 22.9, drift 15 ms $\text{h}^{-1/2}$, slope SD 14.3
PERCLOS	a 3.24%, b 0.62%, σ_e 2.4, κ 1.93	One minute, PERCLOS guard; anchor SD 2.38, drift 0.83% $\text{h}^{-1/2}$, slope SD 0.47

morning of the first test to the third: at about 10:00 (rested), about 03:40 the next night and about 12:00 the next day, rating the KSS at each test. For each test it provides the PVT responses, near-infrared face video from a Kinect sensor, 68 facial landmarks per frame and the vertical electro-oculogram (EOG). Five tests are missing and one did not take place, leaving 36 tests (12 per test number). Video was recorded at about 30 frames per second in all 12 first tests but at about 15 in 8 of the 12 second and 8 of the 12 third tests, so frame rate is confounded with sleep deprivation. Mean KSS rose from 3.25 to 5.58 to 7.33 across the tests.

Signals and sleepiness: Using the eye aspect ratio from the landmarks [19] as the closure score, the website's blink meter gave per-test features whose within-person (repeated-measures [62]) correlations with KSS were 0.77 for PVT median reaction time, 0.73 for lapses (longer than 500 ms, the threshold for the 10-min PVT), 0.53 for PERCLOS, 0.51 for blink duration and 0.35 (not significant) for blink rate (21 degrees of freedom, 13 subjects). Because KSS and every feature also rise with test number, these correlations include the shared time course. The within-person forward-model slopes (Table II) were 18.85 ms of median reaction time, 11.1 ms of blink duration and 0.62 percentage points of PERCLOS per KSS point. The lag-1 autocorrelation of successive minutes was 0.32 for PERCLOS and 0.02 for blink duration, which set κ.

Blink counting against EOG: As a reference we detected blinks in the vertical EOG automatically (band-pass 0.5–15 Hz, prominence at least five scaled median absolute deviations, width 40–600 ms); events were not annotated by hand. Across the 36 tests, the meter's blink rates from the landmarks agreed with EOG blink rates (Spearman $\rho = 0.88$; Pearson $r = 0.89$), with a mean difference of -5.4 blinks min^{-1} and 95% limits of agreement [63] from -19.1 to $+8.3$.

B. The website's camera pipeline on DROZY video

The comparison above tests the blink logic on landmark data, not the pipeline the website runs. We therefore ran the website's MediaPipe model file on every frame of the 36 DROZY videos and passed the mean `eyeBlink` score through the unchanged blink meter (Fig. 3). A face was found in 99.9% of frames (minimum 98.1% in any test).

Blink rates from this pipeline agreed with EOG blink rates across tests (Spearman $\rho = 0.71$) and within person ($r = 0.65$, $p < 0.001$), but undercounted them by 13.0 blinks min^{-1} on average (limits of agreement -35.8 to $+9.8$). Mean blink duration agreed with the landmark pipeline ($\rho = 0.72$) and rose with KSS within person ($r = 0.58$, $p = 0.004$), as did the number of long closures per minute ($r = 0.51$, $p = 0.013$).

PERCLOS failed. In these near-infrared videos MediaPipe's closure score peaked at 0.68 on average (99th percentile per test; mean per-test maximum 0.79), so the normalised closure rarely reached the 0.8 PERCLOS level and PERCLOS was below 0.5% in all 36 tests; it was not related to KSS ($r = 0.22$, $p = 0.32$). When each test's raw score was instead rescaled between its own 5th and 99.5th percentiles, a diagnostic that needs the whole recording, PERCLOS agreed with the landmark PERCLOS ($\rho = 0.86$) and rose with KSS within person ($r = 0.58$, $p = 0.004$), and blink-rate agreement with EOG improved to $\rho = 0.82$. The released model therefore ignores PERCLOS when the camera's blinks do not reach full closure (Section III); an online version of the rescaling is implemented for future validation but is not used. The lag-1 autocorrelation of MediaPipe blink duration was 0.39, against 0.05 for the landmark pipeline computed the same way (0.02 in the forward-model fit, which drops minutes with fewer than three blinks), so the released κ for blink duration is likely too small for the website's pipeline. Restricting to the 20 tests recorded at 30 frames per second gave similar blink-rate agreement with EOG ($\rho = 0.83$), but the within-person association of blink duration with KSS was weaker and not significant ($r = 0.46$, $p = 0.25$; 12 tests, 5 subjects), as was that of long closures ($r = 0.33$, $p = 0.42$). Mean MediaPipe blink duration rose by 84 ms from the first to the second test, three times the 26 ms the forward model predicts for the 2.3-point rise in KSS, and the rise coincides with the drop to 15 frames per second; the blink-duration association is therefore confounded with frame rate.

C. Replay of the released model on DROZY

Design: For each subject in turn, the forward models were refitted on the other 13 subjects. The model then replayed that subject's real test times, PVT median reaction times and per-minute eye features from the landmark pipeline (not the website's MediaPipe pipeline, whose per-minute features were not used here), with or without the KSS rating of the first test as a check-in, and predicted the KSS of each later test from the events before it. DROZY does not record wake time, so we assumed waking at 07:00 after 7.5 h on the day of the first test, with 06:00 and 08:00 as sensitivity analyses. Each rating was placed 11 min after the start of its test. Confidence intervals come from 2000 bootstrap resamples of subjects. Conditions

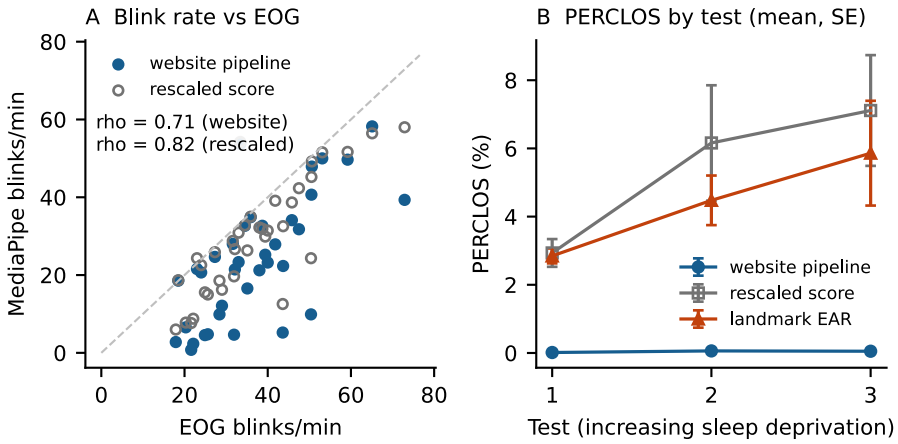


Fig. 3. The website's camera pipeline (MediaPipe `eyeBlink` blendshapes and the unchanged blink meter) run on the 36 DROZY near-infrared videos. (A) Blink rate against EOG blink rate, for the website pipeline and for a diagnostic per-test rescaling of the raw score. (B) Mean PERCLOS by test (SE): the website pipeline stays near zero because the raw score rarely approaches 1.

with a check-in need the first test, so they have 20 predictions from 11 subjects, against 24 from 13 subjects without.

Results: On the 24-prediction set, the mean absolute error (MAE) was 1.73 KSS points for the prior alone, 1.51 for the PVT alone and 1.47 for the PVT and camera features; on the 20-prediction set it was 1.51 for the check-in alone, 1.46 for the check-in and PVT, and 1.38 for all three (Fig. 4). The two sets differ, so these values are not one sequence. The prior over-predicted sleepiness on average (mean error $+0.81$), mostly at the night test (MAE 2.27 at test 2 against 1.18 at test 3); with all sources the mean error fell to $+0.31$ and the test-2 MAE to 1.60. Results were stable to the assumed wake time (all sources: 1.39 at 06:00 and 1.37 at 08:00; prior: 1.75 and 1.71). With two check-ins, predicting the third test (9 subjects), check-ins alone gave 1.90 and all sources 1.78; both under-predicted the final rating.

These results are indicative only. The intervals overlap widely, conditions with and without a check-in differ in their predictions, and two naive baselines bracket the model: carrying the first rating forward gave 3.45, but predicting each test by the mean rating of that test in the other subjects gave 1.39. That baseline knows the protocol, which a person's daily life does not provide, but it shows that the prior's time course, in particular its night-time level, is the largest source of error here. All of the gain over the prior arose at the night test (test 2: 2.27 for the prior, 1.80 for passive signals alone, 1.60 for all sources); at test 3 the passive signals added nothing (1.18, 1.14 and 1.16). Because the forward-model intercepts were learnt on the other DROZY subjects, with the same hardware, PVT and protocol, part of this gain is calibration transfer within one laboratory that corrects the prior's night-time bias, rather than tracking of within-person change; with a browser PVT-B, device latency would shift the reaction-time level by tens of milliseconds, each 18.85 ms being one KSS point. With two check-ins, both methods under-predicted the third test (mean error -1.90 and -1.78), the signature of a night-time error learnt as an offset and carried into the next day. Paired comparisons on common prediction sets, the prior's

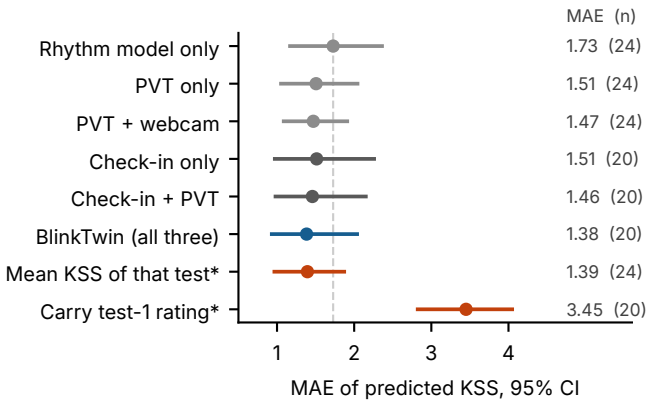


Fig. 4. Leave-one-subject-out replay of the released model on DROZY: MAE of predicted KSS for tests 2 and 3 (95% CI over subjects; n = predictions). The check-in is the rating at test 1. *Naive baselines; the first uses the protocol's average course.

error on the 20-prediction set, and a replay with the website's MediaPipe features need the DROZY recordings and are left to future work.

D. OpenNeuro ds004902

Data: The dataset contains 71 adults tested once after normal sleep and once after a night of sleep deprivation, in counterbalanced order [30], [64]. We used only the participant table (version 1.0.4). KSS and SSS were collected in different participants, and only 2 people have both KSS and PVT data in both sessions.

Group effects and a labelling discrepancy: After sleep deprivation, KSS rose by 2.03 points ($n = 33$, $d_z = 1.00$), the SSS by 1.66 ($n = 35$, $d_z = 0.80$) and the PVT median reaction time by 41.2 ms ($n = 30$, $d_z = 1.33$), while the number of lapses did not change ($p = 0.92$). KSS rose in 73% of people, stayed the same in 18% and fell in 9%. Our paired t statistics for the three PVT columns (0.04, 7.31 and 4.72 in absolute

value for lapses, median and SD of reaction time) equal those in the dataset descriptor, but the descriptor attaches them to different measures. The released data dictionary and the value ranges (0–8 for lapses, 256–423 ms for median reaction time) support our assignment. Across the 25 people with PVT and SSS in both sessions, the change in reaction time was weakly related to the change in SSS (Spearman $\rho = 0.25$, approximate 95% CI -0.17 to 0.59).

Does a person's offset persist?: Anchoring assumes that a person's offset from the prior is stable enough to be worth learning. For the 33 people with KSS in both sessions, we computed each person's offset from the prior in each session (default sleep assumption; KSS time taken as the eyes-open EEG time). The offsets were weakly correlated between sessions ($r = 0.29$, 95% CI -0.05 to 0.58; Spearman 0.32). Predicting the rating after sleep deprivation, the prior alone gave an MAE of 1.53, the prior plus the group's mean offset 1.45, the prior plus the person's shrunk offset from the other session 1.52, the full offset 1.63, and carrying the rating forward 2.33. The group-mean offset predicted best, so in these data the prior was biased for the sample as a whole (the default 07:00 wake time is also an hour earlier than the sample's diary mean of 08:20), and a person's own offset added little. Because the two sessions differ in condition (rested and deprived) as well as in date, this tests a person-by-condition interaction as much as day-to-day persistence. Under the model's random-walk drift, a check-in's mean is carried forward unchanged while its variance grows; a mean-reverting offset, which fades towards a population value, fits these data better and is evaluated in Section VI.

Can the prior be improved?: We fixed two comparisons in advance and report both. First, replacing the default sleep assumption with each person's sleep diary for the normal-sleep night (28 people) did not reduce the prior's MAE over the two sessions (1.74 against 1.69; difference $+0.06$, 95% CI -0.07 to $+0.17$). Second, replacing the KSS mapping $10.6 - 0.6A$ with the mapping refitted to airline data by Ingre et al. [28], $9.68 - 0.46A$, reduced it from 1.69 to 1.46 (difference -0.22, 95% CI -0.33 to -0.09). The default mapping predicted a mean rise of 2.90 points after sleep deprivation and the refitted one 2.23, against an observed 2.04 in these 28 people. With one check-in from the other session, the advantage of the refitted mapping shrank to -0.09 (95% CI -0.26 to $+0.08$; all 33 people). DROZY does not support the same conclusion. There the prior's alertness is almost identical at the night test and the next-noon test (4.66 and 4.63) while the observed mean KSS differs by 1.75 points, which no linear mapping can fit; the refitted mapping lowers the night-test prediction by only 0.27 points, against an over-prediction of about 2.2, and moves the first-test prediction away from the observed value. DROZY's error is in the prior's time course (or the assumed wake time), not in the slope of the mapping. The refitted mapping is therefore a candidate supported by one dataset, and the released default is unchanged.

VI. Simulation Study

A simulation with a known truth shows how much each source of information can contribute and how the model behaves when its assumptions fail. It cannot show that the model works for real people. We report it following Morris et al. [31]; the simulator and the evaluation call the released model code with its default settings, and no parameter was tuned on simulated people.

A. Design

Aims: (i) To measure the information added by check-ins and passive signals; (ii) to test whether estimating per-signal anchors adds anything over a filter that only allows for persistent biases; (iii) to see how results change when the model's assumptions are broken; (iv) to check uncertainty and one-hour forecasts; and (v) to measure sensitivity to the filter settings that are assumptions.

Data-generating mechanisms: Each virtual person has a true latent sleepiness $s^*(t) = \mathrm{clip}\big(p_\phi(t) + o + \gamma^* T(t) + \varepsilon(t)\big)$, where p_ϕ is the Three-Process Model with the person's circadian phase shifted by $\phi \sim \mathcal{N}(0, 1\,\mathrm{h})$, $o \sim \mathcal{N}(0, 0.84)$ is a personal offset (the between-person SD of [28]), $\gamma^* \sim \mathcal{N}(0.4, 0.2)$ is the time-on-task effect, and ε is a slow fluctuation (SD 0.5, time constant 1.5 h). Passive features are generated from person-specific forward models whose population distributions come from DROZY, with slowly drifting webcam baselines and autocorrelated per-minute noise. Ratings are s^* plus noise (SD 0.7), rounded to integers. Each person wakes at 07:00 after 7.5 h of sleep and stays awake for 24 h, alternating 50-min task blocks and 10-min breaks, with a webcam minute every minute from 07:30, a PVT-B every 2 h and 4 min after each check-in. k check-ins are spread from 08:30 to 22:30 ($k = 1$: 10:00). Each person's random numbers do not depend on k, so the truth, the webcam minutes and the 2-hourly reaction tests are identical across k; only the check-ins and the tests that follow them differ. Besides this *matched* scenario, eight scenarios break assumptions: misreported sleep and a circadian phase SD of 2 h; nonlinear signals with heavy-tailed noise; a personal rating bias (SD 0.8) with more rating noise (SD 1.0); strongly autocorrelated webcam noise ($\rho = 0.7$); person-specific homeostatic and circadian parameters (trait vulnerability [65]); five unrecorded nights of 5 h sleep; an ordinary 16-h workday; and ratings that level off above KSS 5 [2]. Details are in Appendix E.

Estimand and methods: The estimand is s^* every 10 min from 07:30 to 07:00 the next day. Methods: the prior alone; check-ins only; the no-anchor and consider-covariance comparators (Section IV); BlinkTwin; and a known-parameter filter, the released filter given each person's true forward-model baselines and slopes. The last uses the model's own noise model, so it is a reference, not an upper bound, and it can be beaten when the generator's noise differs from that model.

Performance measures: Mean absolute error (MAE) per person, averaged over 300 people per scenario, with 95% bootstrap intervals (2000 resamples of people); paired differences with the share of people better; night-time MAE (23:00–07:00); coverage and width of the 90% interval; and, for one-hour forecasts of $s^* \geq 7$ made hourly from 12:00 to 05:00 at $k = 3$, the Brier score [66], the Brier skill score against

TABLE III
SIMULATION, MATCHED SCENARIO: MAE OF LATENT KSS BY NUMBER OF CHECK-INS k (300 PEOPLE), AND COVERAGE AND WIDTH OF THE 90% INTERVAL AT $k = 3$

	MAE by k							$k = 3$	
Method	0	1	2	3	4	6	8	Cov.	Width
Prior alone	0.819							0.98[a]	5.21[a]
Check-ins only		0.744	0.699	0.659	0.628	0.597	0.571	0.91	2.84
No anchors		1.782	1.761	1.741	1.723	1.684	1.655	0.24	1.33
Consider-cov.	1.032	0.791	0.733	0.673	0.637	0.604	0.579	0.90	2.78
BlinkTwin	0.861	0.731	0.667	0.629	0.602	0.581	0.557	0.90	2.64
Known param.	0.724	0.639	0.605	0.586	0.566	0.553	0.536	0.87	2.30

[a] At $k = 0$. Cov.: coverage; width in KSS points.

the prior, the AUROC and the error of the predicted time of reaching KSS 7. All comparisons are exploratory.

B. Results

Check-ins carry most of the information: The prior alone had an MAE of 0.819 KSS points. Three check-ins reduced this to 0.659 and eight to 0.571 (Table III, Fig. 5A).

Passive signals alone did not help: Without check-ins, BlinkTwin's passive signals did not improve on the prior (0.861 against 0.819; difference $+0.042$, 95% CI -0.018 to $+0.099$) and made 41% of people worse by more than 0.1 KSS points (95th percentile of the per-person difference $+0.96$). The known-parameter filter reached 0.724, so the information is there, but it cannot be used without knowing where each person's signals sit. Three check-ins alone also made 29% of people worse than the prior by more than 0.1, although the mean improved.

Estimating the anchors adds a little: With check-ins, BlinkTwin's error was lower than that of check-ins only at every k, with intervals that excluded zero from $k = 2$ (at $k = 3$: -0.030, 95% CI -0.042 to -0.016; 64% of people better, 12% worse by more than 0.1; at night -0.032), with similar coverage (0.90 against 0.91) and narrower intervals (2.64 against 2.84). The consider-covariance filter, which allows for persistent biases without estimating them, was no better than check-ins only ($+0.015$, 95% CI -0.003 to $+0.032$ at $k = 3$; $+0.047$ at $k = 1$) and, without check-ins, worse than the prior ($+0.214$). BlinkTwin beat it at every k (-0.044, 95% CI -0.057 to -0.032 at $k = 3$; 72% of people better). The gain therefore comes from estimating the anchors, and it is small. The no-anchor variant, which treats a persistent bias as independent noise, failed as expected (MAE 1.74, coverage 0.24 at $k = 3$). Coverage fell as check-ins accumulated, from 0.93 at $k = 1$ to 0.87 at $k = 8$ for BlinkTwin (0.95 to 0.88 for check-ins only), because the offset's random walk is the only source of model noise while the truth also fluctuates.

Which signal helps: With three check-ins, the reaction test alone (-0.016, 95% CI -0.024 to -0.007) and the webcam alone (-0.018, -0.026 to -0.009) each carried about half of the gain over check-ins only, and together -0.030 (Table X). Without check-ins, the reaction test alone (0.837) did better than both channels together (0.861); the webcam is what pushed the passive-only estimate above the prior's error. Coverage and width at every k, with Monte Carlo standard errors, are in Table VIII, and per-person quantiles of the gain in Table IX.

When assumptions break: At $k = 3$, BlinkTwin's error was below that of check-ins only in all eight misspecified scenarios (differences -0.013 to -0.055), with an interval that included zero only for strongly autocorrelated webcam noise (Fig. 5B; Table VI); with that noise BlinkTwin was slightly worse than check-ins only at $k = 1$ ($+0.016$) and $k = 8$ ($+0.002$), and the known-parameter filter was worse than BlinkTwin. BlinkTwin was below the consider-covariance filter in all nine scenarios, with an interval that included zero only for biased ratings. When people rated themselves with a personal bias, check-ins anchored the estimate to a biased reference and made it worse than the prior alone (0.918 against 0.819), and BlinkTwin inherited part of that bias (0.863); coverage fell to 0.77. Ratings that level off above KSS 5 changed little, because most check-ins fell before sleepiness rose that far; in that scenario the passive-only estimate, the closest analogue of an objective reading, was worse than check-ins at night (0.871 against 0.624). The unrecorded sleep debt raised the prior's error to 0.858; check-ins reduced it to 0.633 at $k = 3$. On an ordinary workday, the gain over check-ins was -0.021 (95% CI -0.036 to -0.007).

Sensitivity: Across the 13 distinct settings of the six assumed filter parameters (Table VII), BlinkTwin's MAE at $k = 3$ ranged from 0.614 to 0.708 and was never above that of check-ins only. Coverage was most sensitive to the offset drift (0.81 at half the default, 0.96 at double) and to the assumed rating noise (0.82 at $\sigma_K = 0.5$, 0.94 at 1.0). Halving the anchor drift removed the gain over check-ins (-0.002). A mean-reverting offset with a 24-h time constant lowered the error of both methods (BlinkTwin 0.605, check-ins 0.631) without loss of coverage (0.90); with 72 h, coverage fell to 0.83.

Forecasts: All methods projected the protocol's 50/10 work rhythm. In the matched scenario at $k = 3$, 27.5% of one-hour outcomes were $s^* \geq 7$. The Brier score was 0.086 for the prior, 0.066 for check-ins only and 0.063 for BlinkTwin (difference -0.003, 95% CI -0.005 to -0.000), against 0.057 for the known-parameter filter, giving Brier skill scores against the prior of 0.24 and 0.27. The AUROC was 0.945, 0.966 and 0.969 (difference $+0.003$, 95% CI $+0.001$ to $+0.005$), so most of the discrimination comes from the clock. The predicted time of reaching KSS 7, forecast at 17:00, missed by about 80 min with every method (prior 79, check-ins 84, BlinkTwin 81 min, each on the people for whom both forecast and truth crossed: 288, 255 and 257), so neither check-ins nor passive signals improved the timing. The application therefore shows this time with a window, from the first time at which the forecast probability of KSS ≥ 7 reaches 0.25 to the first time it reaches 0.75. The true time fell inside the window for 56% of the people who reached KSS 7 within the horizon, before it for 30% and after it for 15% with BlinkTwin (check-ins only: 56, 30 and 14%), against a nominal 50, 25 and 25%: the window is roughly calibrated but slightly late. Its median width was 130 min with BlinkTwin and 150 min with check-ins only; for the prior alone the probability never reached 0.75 within the horizon. BlinkTwin's Brier score was below

that of check-ins only in every misspecified scenario except the workday, the scenario closest to the intended use, where KSS ≥ 7 occurred in only 1.4% of outcomes and the difference was -0.000 (95% CI -0.001 to $+0.001$); with biased ratings both check-in methods forecast worse than the prior (Brier skill -0.13 and -0.08). Reliability diagrams are in Fig. 8.

VII. Planned Validation Study

The analyses above use public laboratory data and simulation. Whether the approach helps real people in daily life needs a prospective study. We describe the planned design here; before any data are collected it will be registered on OSF Registries with a full statistical analysis plan, under an institutional sponsor and with approval from a human research ethics committee, and only then will it be pre-specified.

Intended use and population: BlinkTwin is intended for self-monitoring by adults during desk work at a laptop. The study recruits adults aged 18–60 who work at a laptop for at least 4 h a day. Exclusions: night-shift work or transmeridian travel within two weeks; diagnosed sleep disorders; a positive screen for obstructive sleep apnoea (STOP-Bang [67]); excessive daytime sleepiness (Epworth Sleepiness Scale above 10 [68]); clinically relevant insomnia (Insomnia Severity Index of 15 or more [69]); pregnancy; and medication that affects alertness. Chronotype is measured with the Morningness–Eveningness Questionnaire [70]. Claims will be limited to this population.

Field phase: Each participant uses BlinkTwin on their own computer for 10 working days in *study mode*, which records events but hides every estimate, forecast, check and suggestion, so that the display cannot anchor ratings or change behaviour. Wrist actigraphy and a sleep diary record sleep; caffeine, alcohol and naps are logged. On each day the participant runs the webcam during laptop work and completes four check-ins at fixed times (09:30, 12:00, 14:30 and 17:00) and two *probe* ratings at random times between them. Probe ratings are withheld from the model and used only for scoring, so that predictions between check-ins, where passive signals can matter, are evaluated. At each check-in the participant rates the KSS first, then takes the 3-min PVT-B, then rates NASA-TLX for the work block just finished. Each device's display and input latency is measured once with a photodiode and a key-press robot, or, where that is not possible, with a browser timing test, and the camera's frame rate is stored with every minute. Participants are told that BlinkTwin must not be used to decide whether they are fit to drive. Bystanders who may appear on the webcam at work, and the employer where relevant, are informed and asked for consent; no video is stored in the field phase.

Laboratory phase: A subset takes part in a supervised session of up to 26 h of continuous wakefulness after at least 7 days of a stable sleep schedule verified by actigraphy, with simultaneous EEG, EOG and video. During the session the camera records desk work (typing, reading, phone use), not only the PVT or resting EEG. Additional exclusions apply (epilepsy, bipolar disorder, cardiovascular disease), caffeine and alcohol are not allowed from 24 h before, a physician is on call, and participants are taken home by arranged transport. Long eye closures and microsleeps are scored from EEG, EOG and video by two blinded raters using the BERN criteria [49], and inter-rater agreement is reported. The number of participants is set by events, not by people: recruitment continues until at least 100 scored long eye closures are observed, which gives a 95% confidence interval half-width of about 0.1 for a sensitivity near 0.7; analysis is per event with clustering by participant. Identifiable video is stored encrypted on institutional servers, used only for scoring and deleted after the study.

Outcomes: Two co-primary outcomes: (1) the absolute error of predicting each probe rating, made from all earlier events (leave-future-out replay of the full 10-day log), with rating noise included in the interval; and (2) the absolute error of predicting the median reaction time of the next PVT-B, an objective outcome. Secondary outcomes: the error of predicting check-in ratings; coverage of the 90% prediction interval, the probability integral transform and the continuous ranked probability score; the Brier score and AUROC for KSS ≥ 7 within the next hour, in the laboratory phase where such ratings are common; sensitivity and positive predictive value of the alerts against scored events (laboratory) and alerts per hour with participant-labelled false alarms (field); agreement of the camera's blink measures with EOG; the false-flag rate of the reactions-versus-ratings check; the error of the predicted next TLX rating (exploratory); usability (System Usability Scale [71]); and adherence.

Hypotheses and analysis: Methods are those of Section VI: the prior alone, check-ins only, the consider-covariance comparator and BlinkTwin. H1: on probe ratings, BlinkTwin is not worse than check-ins only, with a non-inferiority margin of 0.05 KSS points on the per-person mean absolute error (31% of the simulated benefit of check-ins over the prior, 0.16 points), tested with a one-sided 97.5% confidence bound and reported following the CONSORT extension for non-inferiority [72]. H2: on probe ratings, BlinkTwin is better than check-ins only, which is the test of the passive signals. H3: BlinkTwin predicts the next PVT-B better than the prior alone. H1 is tested first; H2 and H3 are tested at a Bonferroni-corrected α of 0.025 each. Absolute errors are also analysed with linear mixed models with method as a fixed effect and random intercepts for participant and day, as a sensitivity analysis. No parameter is tuned on the study data.

Sample size: We estimated power by simulating the analysis that will be run, for check-in ratings: 1000 virtual people (the workday population of Section VI) each had 10 workdays replayed as one continuous event log, so that the offset and anchors carried over between days, with four check-ins a day each followed by a PVT-B; the prediction of each observed, noisy and rounded rating used all earlier events; and studies of n people were drawn from the pool 5000 times. The mean error was 1.01 for the prior alone, 0.85 for check-ins only and 0.84 for BlinkTwin; per person, BlinkTwin minus check-ins averaged -0.010 (SD 0.036) and BlinkTwin minus the consider-covariance comparator -0.014 (SD 0.031). With a mean-reverting offset the difference from check-ins grew to -0.028 (24-h time constant) and -0.047 (72 h). Replaying

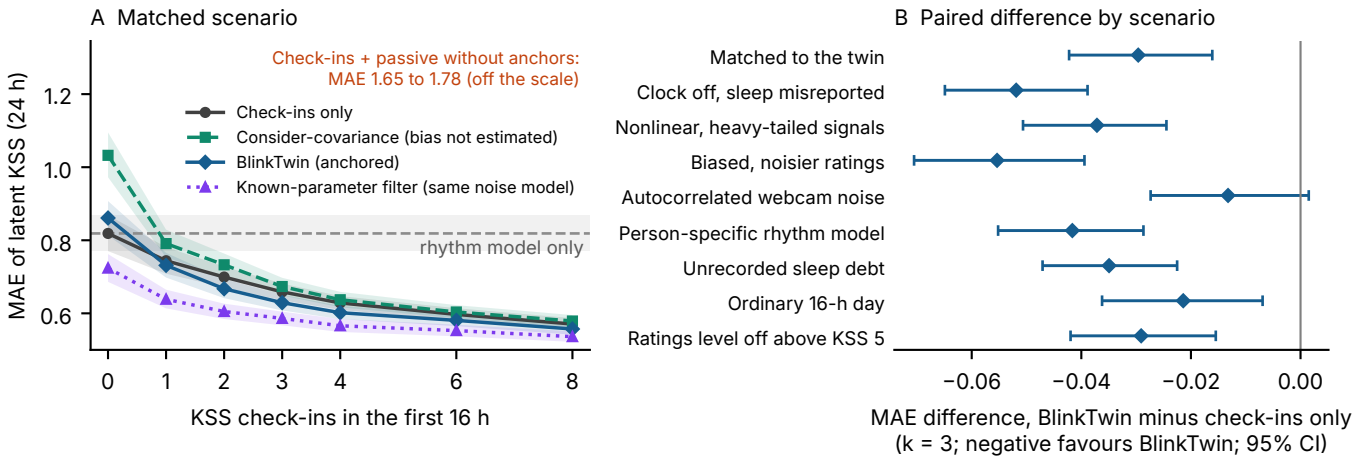


Fig. 5. Simulation (300 virtual people per scenario). (A) Matched scenario: MAE of latent KSS against the number of check-ins, with 95% bootstrap intervals, for check-ins only, the consider-covariance comparator, BlinkTwin and the known-parameter filter; the no-anchor variant is off the scale (MAE 1.65–1.78). (B) Paired difference BlinkTwin minus check-ins only at three check-ins, by scenario (95% CI); negative favours BlinkTwin.

each day separately instead, as in our earlier analysis, reversed the sign ($+0.019$), because the anchors were then relearnt every morning. Non-inferiority at a 0.05 margin had power above 0.99 from 20 people. Superiority of BlinkTwin over check-ins only had power 0.41 with 40, 0.68 with 80, 0.76 with 100 and 0.93 with 150 people (two-sided $\alpha = 0.05$). The co-primary outcomes on probe ratings and the next PVT-B, and the Bonferroni-corrected α, have not yet been simulated; they will be before registration. Provisionally we plan to analyse 120 participants and to recruit 135. The generator keeps each person's offset constant across days, which the weak persistence in ds004902 contradicts, so these figures are likely optimistic for BlinkTwin.

VIII. Discussion

A. Principal findings

On DROZY, point estimates of the error were lower with each added source (MAE 1.73 for the prior alone, 1.38 with a check-in and passive signals), but the intervals overlap, the prediction sets differ, the sample is small, and a protocol-aware naive baseline did as well. All of the gain arose at the night test, where the prior was most biased, and part of it is likely calibration transfer within one laboratory. On ds004902, a group-mean offset predicted ratings better than a person's own offset from another session, and the KSS mapping refitted by Ingre et al. [28] reduced the prior's error by 0.22 points, although DROZY's error lies in the prior's time course rather than its mapping. The website's camera pipeline, run on DROZY's near-infrared video, correlated with EOG blink rate but undercounted blinks by about a third; its blink-duration association with sleepiness was confounded with frame rate; and its PERCLOS failed, so the model now ignores PERCLOS in that case. In simulation, check-ins carried most of the information, and passive signals alone did not improve on the prior. Estimating per-signal anchors gave a small gain (0.03 KSS points over check-ins only, 0.04 over a filter that allows for persistent biases without estimating them), which held when the model's assumptions were broken. In a simulation of the planned 10-day study, replayed as one continuous log, passive signals improved the prediction of observed ratings by 0.01 points, and a mean-reverting offset by a further 0.02 to 0.04; when each day was replayed separately, so that the anchors were relearnt every morning, the passive signals made predictions slightly worse. The value of the anchors therefore depends on carrying them across days. The evidence supports check-ins as the main source of information and anchored passive signals as a small addition whose value in real use is unproven.

B. Comparison with other work

Table I lists published results for related problems. No study we found reports KSS regression error on DROZY or ds004902; published DROZY results are binary classifications [57], which cannot be compared with an MAE, so our numbers are a first reference point on these datasets rather than evidence that BlinkTwin is more accurate than other systems. On the KSS scale, our errors are in the range expected for the Three-Process Model: Ingre et al. reported an in-sample residual SD of 1.46 in field data [28], and Knock et al. predicted 48–60% of shift workers' ratings within one point [55], against 36% for our prior on ds004902 (45% with the refitted mapping). The closest designs predict PVT performance rather than self-rated sleepiness. Subramaniyan et al. fed face-video features into a personalised model and reached an error close to the PVT's own within-person variability [56], the 2B-Alert app needed about 12 PVTs to reach its best accuracy [14], and Ahlstrom et al. improved a sleep/wake predictor's forecast of high KSS with an eye-movement test [53]. That personal learning needs many observations is consistent with our finding that one or two check-ins help little. Accuracy on real drowsiness from consumer cameras is modest: on UTA-RLDD the best video model reached 65.2% on three classes and human judges 57.8% [47].

C. Subjective and objective sleepiness

Anchoring makes the person's ratings the reference, which is right for an estimate of how sleepy a person feels but can hide objective decline [2], [3]. A single reaction test read on the population scale cannot reveal that decline, because people's baselines differ by about 2.6 KSS-equivalents and device latency shifts the level. The application therefore flags a possible dissociation only from within-person change in the reaction-time anchor, and presents the check as experimental. Its false-alarm rate, detection delay and effect on behaviour are untested; a chronic-restriction scenario, in which ratings level off over days while performance declines, is the right test.

D. How accuracy could improve

In order of the evidence we have: (i) a mean-reverting offset, which lowered simulated error without loss of coverage in the one-day study, improved rating prediction most in the 10-day power simulation, and matches the weak persistence in ds004902; (ii) a personal state beyond an additive offset, such as a gain on homeostatic pressure or a circadian phase shift, which trait-like vulnerability [65] predicts and which two sessions at different sleep pressure can identify; (iii) the field-refitted KSS mapping, supported by ds004902 but not by DROZY; (iv) forward models refitted on the website's own camera features, with the closed-eye level normalised to the person, the blink-duration noise inflation matched to that pipeline, long closures as a further signal, and the frame rate, which the application now stores with every minute, used to exclude or model low-rate minutes; (v) a lapse- or speed-based PVT model fitted on the PVT-B itself; and (vi) check-ins scheduled where they are most informative, which the application now suggests.

E. Limitations

The real-data evaluation rests on 14 people in DROZY and 33 in ds004902, with near-infrared rather than RGB video, a 10-min rather than 3-min PVT, assumed wake times in DROZY, frame rate confounded with sleep deprivation, and landmark rather than MediaPipe features in the replay. The forward-model slopes are fitted on noisy, integer ratings and are therefore attenuated [73], and slope heterogeneity is treated as independent noise although it is a persistent person-level error. The EOG reference was an automatic detector without manual annotation, and event-level blink and long-closure accuracy was not measured. Browser reaction times include device latency that we have not yet measured. The KSS is treated as a continuous Gaussian measurement although it is an integer scale with a ceiling, and the model's interval describes the latent state, not a future rating, unless rating noise is added. The prior uses local clock time without phase adaptation, so travel, shift work and daylight-saving changes produce time-varying errors that the offset cannot absorb. The time-on-task term, the offset's dynamics and the alert thresholds are assumptions, and the forward-pitch sign of the head-drop alert has not been verified on devices. The simulation, although it includes misspecified scenarios and a fair comparator, is built from the model's own structure, and the 24-h scenario lies outside the intended use; on an ordinary workday sleepiness of KSS 7 or more was rare. The privacy check used a test pattern without a face. Workload was not evaluated. BlinkTwin is not a medical device.

F. Ethics and privacy

Webcam monitoring at work raises concerns about surveillance. BlinkTwin is designed for self-monitoring: frames never leave the page, a content security policy blocks all connections other than to the site and the model host, only one-minute numeric summaries are stored in the person's browser (unencrypted, which matters on shared computers), and the person can export or delete them. Any deployment by an employer would need informed consent, transparency about what is measured, and a guarantee that the data are not used against the person.

IX. Conclusion

BlinkTwin estimates how sleepy a person feels by joining the published Three-Process Model with occasional KSS check-ins and with reaction-time and camera signals that each have their own personal anchor, entirely in a browser and with video kept on the device. On public data, check-ins and the prior's time course mattered most, the camera pipeline measured blinks but not PERCLOS on near-infrared video, and the passive signals' benefit was small and confined to simulation. Validation of the camera pipeline on ordinary webcams and the planned prospective study are needed before any claim about daily use.

Appendix A
Workload model

BlinkTwin also keeps an estimate of perceived workload $w(t)$ on the raw NASA-TLX scale (0–100) [58], [74]. NASA-TLX rates a completed task, so treating a rating as a point observation of a momentary state is an approximation; the check-in therefore asks about the work block just finished. Workload reverts towards the demand μ_c of the current context c as an Ornstein–Uhlenbeck process with time constant $\tau_w = 20$ min and stationary SD $\sigma_\infty = 18$. Over a step of Δ hours, with $\beta = e^{-60\Delta/\tau_w}$,

$$w \leftarrow \mu_c + (w - \mu_c)\,\beta, \qquad P_w \leftarrow P_w\,\beta^2 + \sigma_\infty^2\,(1-\beta^2). \quad (14)$$

The demand of each context starts from a default D_c (break 15, reading 40, focused work 65, meeting 50, other or unset 40) and is shrunk towards the person's own TLX ratings in that context, $\mu_c = (2D_c + \sum_i y_{c,i})/(2 + n_c)$. A TLX rating updates w with noise SD $\sigma_T = 8$; the initial state is 40 with SD 20. The webcam is not used for workload. The defaults are assumptions, the same rating enters both the learned demand and the update, and the component has not been evaluated; it is therefore described as a feature of the application, not as a validated estimate of cognitive load [75].

APPENDIX B
THREE-PROCESS MODEL CONSTANTS

Table IV lists every constant of the prior with its source. Each value was checked against [6] and [28]. The linear recovery rate below the breakpoint is the exponential's slope at the breakpoint, $\delta_s(H - B) = 0.80$ units h^{-1}, following the construction in [6]. Sleep inertia uses the values of [28]; the 2004 paper lists $W_{\mathrm{Decay}} = 0.65692$, which equals 1.51 divided by $\ln 10$ and is therefore probably a base-10 rate. Ingre et al. found that adding W worsened the fit to airline data and refitted the KSS mapping to $a_K = 9.68$, $b_K = 0.46$ [28]; Section V-D tests that mapping on OpenNeuro ds004902.

TABLE IV
THREE-PROCESS MODEL CONSTANTS AND SOURCES

Constant	Value	Source
Lower, upper asymptote of S	2.4, 14.3	[6], [28]
Decay of S awake	0.0353 h^{-1}	[6], [28]
Recovery of S asleep	0.381 h^{-1}	[6], [28]
Breakpoint	12.2	[27], [28]
Circadian amplitude, acrophase	2.5, 16.8 h	[6], [28]
Ultradian amplitude, mesor	0.5, -0.5	[28]
Ultradian dip	13.8 h (and 01.8 h)	[6]
Sleep inertia at waking, decay	-5.72, 1.51 h^{-1}	[28]
KSS mapping	$10.6 - 0.6A$	[28]
Habitual nights before history	10×7.5 h	this work

APPENDIX C
DROZY: DESCRIPTIVE RESULTS

Table V gives the means of every measure by test, for the landmark pipeline, the website's MediaPipe pipeline and the EOG; only aggregates are shown, following the dataset's licence. Fig. 6 shows the agreement of the landmark blink rates with EOG. In random-intercept mixed models, KSS rose by 2.22 points per 100 ms of median reaction time (95% CI 1.07 to 3.37), 0.28 points per percentage point of PERCLOS (0.08 to 0.48) and 1.92 points per 100 ms of blink duration (0.71 to 3.14); these Wald intervals are anti-conservative with 14 clusters, the models do not separate within- from between-person effects, and the blink-rate model did not converge. They are reported for comparison with the earlier draft; the released model uses the within-person forward models of the main text instead. Per-subject slopes of KSS on reaction time, for the 9 subjects with three tests, ranged from 2.5 to 7.5 points per 100 ms, all positive; with three points per subject their spread cannot be separated from estimation noise.

Replay details: In each fold of the leave-one-subject-out replay (Section V-C, `web/scripts/drozy_twin.ts`), three numbers per signal were refitted on the other 13 subjects with the within-person regression of the main text: the slope b, the population level a (the mean of the subjects' intercepts) and the between-person SD of the anchor (the spread of the intercepts less its share of measurement noise). Everything else kept its released value: the measurement-noise SDs and their inflation for serial correlation, the slope spread, the anchor drift, the offset's prior and drift, the time-on-task term and all Three-Process Model constants. Times were the local clock times in the PVT files, used without a time-zone shift. The test's PVT result entered 10 min after the test started, at the end of the 10-min test; each minute of eye features entered at the end of its minute, and blink duration only from minutes with at least three blinks. DROZY records no time for the KSS rating, so each rating was placed 11 min after the test started, and each prediction was made just before it, from all earlier events: the same test's PVT and eye features were therefore used, its rating was not. The landmark features carry no blink-peak value, so the PERCLOS guard of Section III did not apply and PERCLOS was always used.

TABLE V
DROZY: MEAN (SD) OF EACH MEASURE BY TEST (12 TESTS EACH)

Measure	Test 1	Test 2	Test 3
KSS	3.25 (1.29)	5.58 (1.83)	7.33 (1.37)
PVT median RT (ms)	346 (38)	391 (83)	417 (83)
PVT lapses >500 ms	4.3 (4.3)	14.8 (18.1)	23.8 (18.3)
Landmark (eye aspect ratio) pipeline			
PERCLOS (%)	2.85 (0.76)	4.48 (2.52)	5.86 (5.32)
Blink rate (min^{-1})	32.0 (16.8)	31.6 (16.2)	31.1 (14.4)
Blink duration (ms)	180 (47)	225 (55)	243 (50)
Long closures (min^{-1})	0.17 (0.38)	0.69 (0.98)	1.36 (2.13)
Website pipeline (MediaPipe), mean only			
PERCLOS (%)	0.01	0.06	0.05
Blink rate (min^{-1})	22.4	24.8	24.7
Blink duration (ms)	159	242	226
Long closures (min^{-1})	0.02	0.41	0.89
EOG blink rate (min^{-1})	35.9	37.0	38.0
Tests at about 15 fps	0 of 12	8 of 12	8 of 12

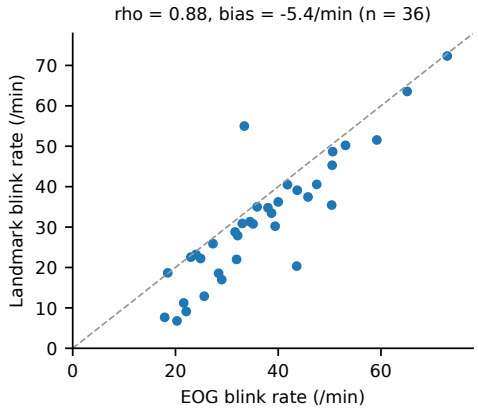


Fig. 6. Blink rate from the website's blink meter, run on DROZY landmarks, against blink rate from the vertical EOG, one point per test ($n = 36$).

APPENDIX D
OPENNEURO DS004902: DETAILS

Fig. 7 shows the KSS and PVT median reaction time in the two sessions and the coupling of the reaction-time change with the SSS change. For the offset analysis, the KSS time was the eyes-open EEG time of each session; the normal-sleep session assumed a 7.5-h night ending at 07:00 and the deprivation session assumed waking at 07:00 on the previous day. The sleep-diary analysis used the diary's bedtime and get-up time for the normal-sleep night (28 people with KSS in both sessions and a diary between 3 and 13 h), and the diary's

get-up time as the previous wake time for the deprivation session. The shrinkage factor for one check-in was the Kalman gain $\sigma_d^2/(\sigma_d^2+\sigma_K^2)=0.72$; in the persistence analysis it was estimated from the data as 0.77.

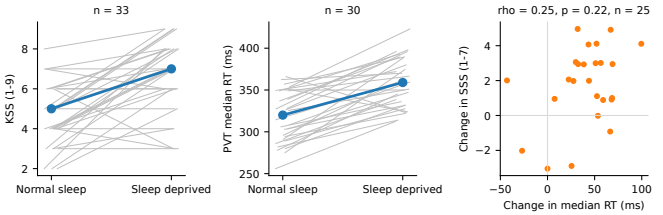


Fig. 7. OpenNeuro ds004902. Left and middle: KSS and PVT median reaction time after normal sleep and after sleep deprivation (grey: individuals; blue: medians). Right: change in median reaction time against change in SSS for the 25 people with both.

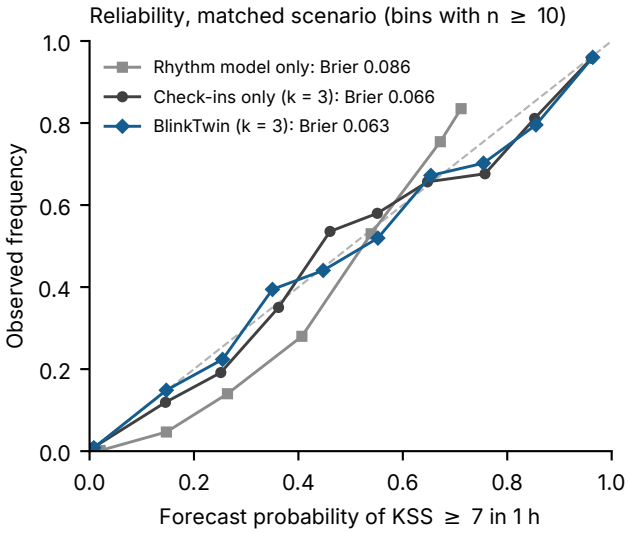


Fig. 8. Reliability of the one-hour forecast of latent KSS ≥ 7 in the matched scenario at three check-ins (10 equal-width bins; 5400 forecasts from 300 virtual people). Points on the diagonal are perfectly calibrated.

Appendix E
Simulation Details

Virtual people: Person-level parameters: offset $o \sim \mathcal{N}(0, 0.84)$; circadian phase shift $\mathcal{N}(0,1)$ h; time-on-task effect $\mathcal{N}(0.4, 0.2)$ KSS h^{-1}, truncated at 0; slow fluctuation of sleepiness (Ornstein–Uhlenbeck, SD 0.5, time constant 1.5 h); rating noise SD 0.7, rounded and clipped to 1–9. Each passive feature follows $a_i + b_i(s^* - 3)$ with $a_i \sim \mathcal{N}(a, \sigma_\alpha)$ and $b_i \sim \mathcal{N}(b, \cdot)$ truncated at 0, using the DROZY means of Table II but smaller slope spreads (SD 8 ms, 7 ms and 0.3% for reaction time, blink duration and PERCLOS) than the filter assumes (7.9 ms, 14.3 ms and 0.47%). The filter's values are the raw spread of per-person slopes from the 8 subjects with three usable sessions, an upper bound that includes estimation noise; the simulator's values are a judgement of the true spread. The matched scenario therefore does not test a filter that underestimates slope heterogeneity. Webcam baselines drift (Ornstein–Uhlenbeck, SD 30 ms for blink duration and 1.8% for PERCLOS, time constant 4 h) and per-minute noise is AR(1) with $\rho = 0.02$ (blink duration) and 0.32 (PERCLOS). Truncations: median reaction time ≥ 180 ms; blink duration ≥ 60 ms; PERCLOS in 0–60%; blink rate $\mathcal{N}(18, 6)$ per minute, rounded, with blink duration missing below 3 blinks; face coverage fixed at 0.9.

Scenarios: Misreport: circadian phase SD 2 h and actual sleep differing from the reported 7.5 h with SD 1.5 h. *Nonlinear*: signals respond with 0.6 of the slope up to KSS 6 and an added quadratic term above it; Student-t_3 noise. *Rating bias*: a personal bias SD 0.8 added to every rating and rating noise SD 1.0. *Autocorrelated webcam*: per-minute noise AR(1) with $\rho = 0.7$ and larger baseline drift. *Trait*: homeostatic decay multiplied by $e^{\mathcal{N}(0,0.3)}$ and circadian amplitude by $e^{\mathcal{N}(0,0.2)}$ per person. *Debt*: five previous nights of 5 h, of which the model is told only the last. *Workday*: 16 h awake (07:30–23:00), check-ins 08:30–21:30, forecasts 09:00–22:00. *Dissociation*: ratings equal the truth up to KSS 5 and include only half of any further rise.

Data and Code Availability

The source code of BlinkTwin, the analysis scripts, the simulation code, the result files and the scripts that produce every figure and table are at https://github.com/r11up/BlinkTwin; the version described here is tagged `v1.0-arxiv`, and the code is released under the MIT licence. The application runs in a current browser with no installation. The TypeScript scripts run directly in Node.js 22.6 or later (type stripping), and the Python analyses use Python 3.13 with pinned packages; the repository's README lists the commands that reproduce every number. OpenNeuro ds004902 is openly available [64]. DROZY is available from the University of Liège under a research-only licence that does not permit redistribution [29]; it is not included, a script downloads it for users who accept its terms, and only aggregate results derived from it are reported or stored in the repository.

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References

[1] D. Dawson and K. Reid, "Fatigue, alcohol and performance impairment," *Nature*, vol. 388, no. 6639, p. 235, 1997.

[2] H. P. A. Van Dongen, G. Maislin, J. M. Mullington, and D. F. Dinges, "The cumulative cost of additional wakefulness: Dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restriction and total sleep deprivation," *Sleep*, vol. 26, no. 2, pp. 117–126, 2003.

[3] R. Leproult, E. F. Colecchia, A. M. Berardi, R. Stickgold, S. M. Kosslyn, and E. Van Cauter, "Individual differences in subjective and objective alertness during sleep deprivation are stable and unrelated," *American Journal of Physiology. Regulatory, Integrative and Comparative Physiology*, vol. 284, no. 2, pp. R280–R290, 2003.

TABLE VI
MAE OF LATENT KSS AT THREE CHECK-INS BY SCENARIO, AND PAIRED DIFFERENCES OF BLINKTWIN AGAINST CHECK-INS ONLY (WITH THE SHARE OF PEOPLE FOR WHOM BLINKTWIN WAS BETTER) AND AGAINST THE CONSIDER-COVARIANCE COMPARATOR (95% CI)

Scenario	Prior	Check-ins	No anchors	Consider	BlinkTwin	Known	BlinkTwin − check-ins (95% CI)	Better	BlinkTwin − consider (95% CI)
Matched	0.819	0.659	1.741	0.673	0.629	0.586	−0.030 (−0.042, −0.016)	64%	−0.044 (−0.057, −0.032)
Misreport	0.924	0.763	1.735	0.763	0.711	0.660	−0.052 (−0.065, −0.039)	66%	−0.052 (−0.064, −0.040)
Nonlinear	0.837	0.671	1.754	0.669	0.633	0.615	−0.037 (−0.051, −0.024)	63%	−0.036 (−0.048, −0.025)
Rating bias	0.819	0.918	1.742	0.875	0.863	0.691	−0.055 (−0.071, −0.039)	68%	−0.012 (−0.027, +0.003)
Autocorrelated	0.819	0.659	2.026	0.693	0.646	0.657	−0.013 (−0.027, +0.001)	53%	−0.048 (−0.062, −0.033)
Trait	0.969	0.697	1.728	0.706	0.656	0.606	−0.042 (−0.055, −0.029)	66%	−0.051 (−0.063, −0.038)
Debt	0.858	0.633	1.768	0.642	0.598	0.557	−0.035 (−0.047, −0.023)	65%	−0.044 (−0.055, −0.033)
Workday	0.832	0.682	1.732	0.701	0.661	0.640	−0.021 (−0.036, −0.007)	56%	−0.040 (−0.053, −0.027)
Dissociation	0.819	0.665	1.742	0.676	0.636	0.585	−0.029 (−0.042, −0.015)	62%	−0.041 (−0.053, −0.027)

TABLE VII
SENSITIVITY OF THE MATCHED SCENARIO AT THREE CHECK-INS TO THE ASSUMED FILTER SETTINGS (DEFAULT IN BOLD)

Setting	Value	MAE		Coverage	
		Check-ins	BlinkTwin	Check-ins	BlinkTwin
Offset prior SD σ_d	0.84	0.647	0.614	0.90	0.90
	1.2	0.659	0.629	0.91	0.90
	1.6	0.671	0.647	0.91	0.90
Offset drift $\sqrt{q_d}$	0.15	0.629	0.615	0.82	0.81
	0.3	0.659	0.629	0.91	0.90
	0.6	0.720	0.708	0.98	0.96
Time on task γ	0	0.653	0.620	0.91	0.91
	0.4	0.659	0.629	0.91	0.90
	0.8	0.700	0.669	0.88	0.88
Rating noise σ_K	0.5	0.707	0.668	0.82	0.82
	0.75	0.659	0.629	0.91	0.90
	1.0	0.630	0.618	0.95	0.94
Anchor drift (×)	0.5	0.659	0.657	0.91	0.87
	1	0.659	0.629	0.91	0.90
	2	0.659	0.635	0.91	0.90
Slope spread (×)	0	0.659	0.633	0.91	0.90
	1	0.659	0.629	0.91	0.90
	2	0.659	0.628	0.91	0.90
Mean-reverting τ_d	24 h	0.631	0.605	0.90	0.90
	72 h	0.622	0.606	0.84	0.83

TABLE VIII
MATCHED SCENARIO: MAE WITH ITS MONTE CARLO STANDARD ERROR (APPROXIMATED AS THE 95% BOOTSTRAP INTERVAL WIDTH DIVIDED BY 3.92), AND COVERAGE AND WIDTH OF THE 90% INTERVAL, AT EVERY NUMBER OF CHECK-INS k

Method	k	MAE	MC SE	Coverage	Width
Prior alone	0	0.819	0.025	0.984	5.206
Check-ins only	1	0.744	0.019	0.945	3.818
	2	0.699	0.015	0.928	3.218
	3	0.659	0.012	0.906	2.840
	4	0.628	0.010	0.899	2.650
	6	0.597	0.008	0.880	2.424
	8	0.571	0.008	0.876	2.306
No anchors	1	1.782	0.047	0.237	1.349
	2	1.761	0.045	0.238	1.337
	3	1.741	0.046	0.241	1.330
	4	1.723	0.046	0.242	1.323
	6	1.684	0.045	0.245	1.309
	8	1.655	0.042	0.246	1.300
Consider-covariance	0	1.032	0.031	0.927	4.646
	1	0.791	0.019	0.924	3.677
	2	0.733	0.016	0.905	3.132
	3	0.673	0.012	0.896	2.780
	4	0.637	0.011	0.892	2.600
	6	0.604	0.009	0.876	2.383
	8	0.579	0.008	0.869	2.269
BlinkTwin	0	0.861	0.024	0.952	4.253
	1	0.731	0.017	0.929	3.362
	2	0.667	0.014	0.917	2.946
	3	0.629	0.011	0.901	2.642
	4	0.602	0.009	0.895	2.476
	6	0.581	0.008	0.875	2.273
	8	0.557	0.008	0.871	2.164
Known parameters	0	0.724	0.019	0.867	2.862
	1	0.639	0.013	0.886	2.658
	2	0.605	0.011	0.883	2.457
	3	0.586	0.010	0.872	2.295
	4	0.566	0.009	0.868	2.192
	6	0.553	0.008	0.852	2.050
	8	0.536	0.007	0.846	1.968

[4] A. A. Borbély, "A two process model of sleep regulation," *Human Neurobiology*, vol. 1, no. 3, pp. 195–204, 1982.

[5] T. Åkerstedt and S. Folkard, "The three-process model of alertness and its extension to performance, sleep latency, and sleep length," *Chronobiology International*, vol. 14, no. 2, pp. 115–123, 1997.

[6] T. Åkerstedt, S. Folkard, and C. Portin, "Predictions from the three-process model of alertness," *Aviation, Space, and Environmental Medicine*, vol. 75, no. 3 Suppl, pp. A75–A83, 2004.

[7] S. R. Hursh, D. P. Redmond, M. L. Johnson *et al.*, "Fatigue models for applied research in warfighting," *Aviation, Space, and Environmental Medicine*, vol. 75, no. 3 Suppl, pp. A44–A53, 2004.

[8] M. M. Mallis, S. Mejdal, T. T. Nguyen, and D. F. Dinges, "Summary of the key features of seven biomathematical models of human fatigue and performance," *Aviation, Space, and Environmental Medicine*, vol. 75, no. 3 Suppl, pp. A4–A14, 2004.

[9] D. Dawson, Y. I. Noy, M. Härmä, T. Åkerstedt, and G. Belenky, "Modelling fatigue and the use of fatigue models in work settings," *Accident Analysis & Prevention*, vol. 43, no. 2, pp. 549–564, 2011.

[10] E. Olofsen, D. F. Dinges, and H. P. A. Van Dongen, "Nonlinear mixed-effects modeling: Individualization and prediction," *Aviation, Space, and Environmental Medicine*, vol. 75, no. 3 Suppl, pp. A134–A140, 2004.

[11] H. P. A. Van Dongen, C. G. Mott, J.-K. Huang, D. J. Mollicone, F. D. McKenzie, and D. F. Dinges, "Optimization of biomathematical model predictions for cognitive performance impairment in individuals: Accounting for unknown traits and uncertain states in homeostatic and circadian processes," *Sleep*, vol. 30, no. 9, pp. 1129–1143, 2007.

[12] S. Rajaraman, A. V. Gribok, N. J. Wesensten, T. J. Balkin, and J. Reifman, "Individualized performance prediction of sleep-deprived individuals with the two-process model," *Journal of Applied Physiology*, vol. 104, no. 2, pp. 459–468, 2008.

[13] J. Liu, S. Ramakrishnan, S. Laxminarayan, T. J. Balkin, and J. Reifman, "Real-time individualization of the unified model of performance," *Journal of Sleep Research*, vol. 26, no. 6, pp. 820–831, 2017.

[14] J. Reifman, S. Ramakrishnan, J. Liu, A. Kapela, T. J. Doty, T. J. Balkin, K. Kumar, and M. Y. Khitrov, "2B-Alert App: A mobile application for real-time individualized prediction of alertness," *Journal of Sleep Research*, vol. 28, no. 2, p. e12725, 2019.

[15] W. W. Wierwille *et al.*, "Research on vehicle-based driver status/performance monitoring; development, validation, and refinement of algorithms for detection of driver drowsiness," National Highway Traffic Safety Administration, Washington, DC, Final Report DOT HS 808 247,

TABLE IX
PER-PERSON DIFFERENCES, BLINKTWIN MINUS CHECK-INS ONLY, AT THREE CHECK-INS: MEAN WITH 95% INTERVAL, 5TH, 50TH AND 95TH PERCENTILES, SHARE OF PEOPLE BETTER, AND SHARE WORSE BY MORE THAN 0.1 KSS POINTS

Scenario	Mean	95% CI	p05	p50	p95	Better	Worse > 0.1
Matched	-0.030	-0.042, -0.016	-0.207	-0.027	0.161	0.640	0.123
Misreport	-0.052	-0.065, -0.039	-0.243	-0.056	0.149	0.663	0.110
Nonlinear	-0.037	-0.051, -0.024	-0.218	-0.036	0.133	0.627	0.083
Rating bias	-0.055	-0.071, -0.039	-0.273	-0.054	0.161	0.683	0.123
Autocorrelated	-0.013	-0.027, 0.001	-0.210	-0.011	0.187	0.527	0.163
Trait	-0.042	-0.055, -0.029	-0.230	-0.048	0.160	0.663	0.120
Debt	-0.035	-0.047, -0.023	-0.220	-0.043	0.151	0.650	0.103
Workday	-0.021	-0.036, -0.007	-0.233	-0.022	0.174	0.563	0.133
Dissociation	-0.029	-0.042, -0.015	-0.207	-0.032	0.167	0.620	0.133

TABLE X
CHANNEL ABLATION, MATCHED SCENARIO: MAE OF LATENT KSS (95% INTERVAL) FOR BLINKTWIN WITH EACH PASSIVE CHANNEL ALONE

Method	$k = 0$	$k = 1$	$k = 3$	$k = 8$
Check-ins only	–	0.744	0.659	0.571
Reaction test only	0.837	0.737	0.643	0.564
Webcam only	0.864	0.732	0.641	0.560
Both (BlinkTwin)	0.861	0.731	0.629	0.557
Paired difference from check-ins only at $k = 3$ (95% interval)				
Reaction test only	-0.016 (-0.024, -0.007)			
Webcam only	-0.018 (-0.026, -0.009)			
Both	-0.030 (-0.042, -0.016)			

1994.

[16] D. F. Dinges, M. M. Mallis, G. Maislin, and J. W. Powell, "Evaluation of techniques for ocular measurement as an index of fatigue and as the basis for alertness management," National Highway Traffic Safety Administration, Washington, DC, Final Report DOT HS 808 762, 1998.

[17] P. P. Caffier, U. Erdmann, and P. Ullsperger, "Experimental evaluation of eye-blink parameters as a drowsiness measure," *European Journal of Applied Physiology*, vol. 89, no. 3-4, pp. 319–325, 2003.

[18] R. Schleicher, N. Galley, S. Briest, and L. Galley, "Blinks and saccades as indicators of fatigue in sleepiness warnings: Looking tired?" *Ergonomics*, vol. 51, no. 7, pp. 982–1010, 2008.

[19] T. Soukupová and J. Čech, "Real-time eye blink detection using facial landmarks," in *21st Computer Vision Winter Workshop (CVWW)*, Rimske Toplice, Slovenia, 2016.

[20] M. Ingre, T. Åkerstedt, B. Peters, A. Anund, and G. Kecklund, "Subjective sleepiness, simulated driving performance and blink duration: Examining individual differences," *Journal of Sleep Research*, vol. 15, no. 1, pp. 47–53, 2006.

[21] T. Åkerstedt and M. Gillberg, "Subjective and objective sleepiness in the active individual," *International Journal of Neuroscience*, vol. 52, no. 1-2, pp. 29–37, 1990.

[22] K. Kaida, M. Takahashi, T. Åkerstedt, A. Nakata, Y. Otsuka, T. Haratani, and K. Fukasawa, "Validation of the Karolinska sleepiness scale against performance and EEG variables," *Clinical Neurophysiology*, vol. 117, no. 7, pp. 1574–1581, 2006.

[23] T. Åkerstedt, A. Anund, J. Axelsson, and G. Kecklund, "Subjective sleepiness is a sensitive indicator of insufficient sleep and impaired waking function," *Journal of Sleep Research*, vol. 23, no. 3, pp. 240–252, 2014.

[24] M. Grieves and J. Vickers, "Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems," in *Transdisciplinary Perspectives on Complex Systems*, F.-J. Kahlen, S. Flumerfelt, and A. Alves, Eds. Cham: Springer, 2017, pp. 85–113.

[25] J. Corral-Acero, F. Margara, M. Marciniak *et al.*, "The 'digital twin' to enable the vision of precision cardiology," *European Heart Journal*, vol. 41, no. 48, pp. 4556–4564, 2020.

[26] B. Friedland, "Treatment of bias in recursive filtering," *IEEE Transactions on Automatic Control*, vol. 14, no. 4, pp. 359–367, 1969.

[27] T. Åkerstedt, M. Ingre, G. Kecklund, S. Folkard, and J. Axelsson, "Accounting for partial sleep deprivation and cumulative sleepiness in the three-process model of alertness regulation," *Chronobiology International*, vol. 25, no. 2, pp. 309–319, 2008.

[28] M. Ingre, W. Van Leeuwen, T. Klemets, C. Ullvetter, S. Hough, G. Kecklund, D. Karlsson, and T. Åkerstedt, "Validating and extending the three process model of alertness in airline operations," *PLoS ONE*, vol. 9, no. 10, p. e108679, 2014.

[29] Q. Massoz, T. Langohr, C. François, and J. G. Verly, "The ULg multimodality drowsiness database (called DROZY) and examples of use," in *2016 IEEE Winter Conference on Applications of Computer Vision (WACV)*, Lake Placid, NY, USA, 2016, pp. 1–7.

[30] C. Xiang, X. Fan, D. Bai, K. Lv, and X. Lei, "A resting-state EEG dataset for sleep deprivation," *Scientific Data*, vol. 11, p. 427, 2024.

[31] T. P. Morris, I. R. White, and M. J. Crowther, "Using simulation studies to evaluate statistical methods," *Statistics in Medicine*, vol. 38, no. 11, pp. 2074–2102, 2019.

[32] S. F. Schmidt, "Application of state-space methods to navigation problems," *Advances in Control Systems*, vol. 3, pp. 293–340, 1966.

[33] A. Å. Miley, G. Kecklund, and T. Åkerstedt, "Comparing two versions of the Karolinska Sleepiness Scale (KSS)," *Sleep and Biological Rhythms*, vol. 14, no. 3, pp. 257–260, 2016.

[34] E. Hoddes, V. Zarcone, H. Smythe, R. Phillips, and W. C. Dement, "Quantification of sleepiness: A new approach," *Psychophysiology*, vol. 10, no. 4, pp. 431–436, 1973.

[35] D. F. Dinges and J. W. Powell, "Microcomputer analyses of performance on a portable, simple visual RT task during sustained operations," *Behavior Research Methods, Instruments, & Computers*, vol. 17, no. 6, pp. 652–655, 1985.

[36] M. Basner and D. F. Dinges, "Maximizing sensitivity of the psychomotor vigilance test (PVT) to sleep loss," *Sleep*, vol. 34, no. 5, pp. 581–591, 2011.

[37] M. Basner, D. Mollicone, and D. F. Dinges, "Validity and sensitivity of a brief psychomotor vigilance test (PVT-B) to total and partial sleep deprivation," *Acta Astronautica*, vol. 69, no. 11-12, pp. 949–959, 2011.

[38] D. A. Grant, K. A. Honn, M. E. Layton, S. M. Riedy, and H. P. A. Van Dongen, "3-minute smartphone-based and tablet-based psychomotor vigilance tests for the assessment of reduced alertness due to sleep deprivation," *Behavior Research Methods*, vol. 49, no. 3, pp. 1020–1029, 2017.

[39] A. Anwyl-Irvine, E. S. Dalmaijer, N. Hodges, and J. K. Evershed, "Realistic precision and accuracy of online experiment platforms, web browsers, and devices," *Behavior Research Methods*, vol. 53, no. 4, pp. 1407–1425, 2021.

[40] D. Bridges, A. Pitiot, M. R. MacAskill, and J. W. Peirce, "The timing mega-study: Comparing a range of experiment generators, both lab-based and online," *PeerJ*, vol. 8, p. e9414, 2020.

[41] D. F. Dinges and R. Grace, "PERCLOS: A valid psychophysiological measure of alertness as assessed by psychomotor vigilance," Federal Highway Administration, Office of Motor Carriers, Tech Brief FHWA-MCRT-98-006, 1998.

[42] V. E. Wilkinson, M. L. Jackson, J. Westlake, B. Stevens, M. Barnes, P. Swann, S. M. W. Rajaratnam, and M. E. Howard, "The accuracy of eyelid movement parameters for drowsiness detection," *Journal of Clinical Sleep Medicine*, vol. 9, no. 12, pp. 1315–1324, 2013.

[43] C. Anderson, A.-M. Chang, J. P. Sullivan, J. M. Ronda, and C. A. Czeisler, "Assessment of drowsiness based on ocular parameters detected by infrared reflectance oculography," *Journal of Clinical Sleep Medicine*, vol. 9, no. 9, pp. 907–920, 2013.

[44] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, "Real-time facial surface geometry from monocular video on mobile GPUs," arXiv:1907.06724, 2019.

[45] C. Lugaresi, J. Tang, H. Nash *et al.*, "MediaPipe: A framework for building perception pipelines," arXiv:1906.08172, 2019.

[46] I. Grishchenko *et al.*, "Blendshapes GHUM: Real-time monocular facial blendshape prediction," arXiv:2309.05782, 2023.

[47] R. Ghoddoosian, M. Galib, and V. Athitsos, "A realistic dataset and baseline temporal model for early drowsiness detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019, pp. 178–187.

[48] Q. Massoz, J. G. Verly, and M. Van Droogenbroeck, "Multi-timescale drowsiness characterization based on a video of a driver's face," *Sensors*, vol. 18, no. 9, p. 2801, 2018.

[49] A. Hertig-Godeschalk, J. Skorucak, A. Malafeev, P. Achermann, J. Mathis, and D. R. Schreier, "Microsleep episodes in the borderland between wakefulness and sleep," *Sleep*, vol. 43, no. 1, p. zsz163, 2020.

[50] T. Åkerstedt and S. Folkard, "Validation of the S and C components of the three-process model of alertness regulation," *Sleep*, vol. 18, no. 1, pp. 1–6, 1995.

[51] S. Ramakrishnan, N. J. Wesensten, G. H. Kamimori, J. E. Moon, T. J. Balkin, and J. Reifman, "A unified model of performance for predicting the effects of sleep and caffeine," *Sleep*, vol. 39, no. 10, pp. 1827–1841, 2016.

[52] D. Sandberg, T. Åkerstedt, A. Anund, G. Kecklund, and M. Wahde, "Detecting driver sleepiness using optimized nonlinear combinations of sleepiness indicators," *IEEE Transactions on Intelligent Transportation Systems*, vol. 12, no. 1, pp. 97–108, 2011.

[53] C. Ahlstrom, M. Nyström, K. Holmqvist, C. Fors, D. Sandberg, A. Anund, G. Kecklund, and T. Åkerstedt, "Fit-for-duty test for estimation of drivers' sleepiness level: Eye movements improve the sleep/wake predictor," *Transportation Research Part C: Emerging Technologies*, vol. 26, pp. 20–32, 2013.

[54] T. Asparouhov, E. L. Hamaker, and B. Muthén, "Dynamic structural equation models," *Structural Equation Modeling: A Multidisciplinary Journal*, vol. 25, no. 3, pp. 359–388, 2018.

[55] S. A. Knock *et al.*, "Prediction of shiftworker alertness, sleep, and circadian phase using a model of arousal dynamics constrained by shift schedules and light exposure," *Sleep*, vol. 44, no. 11, p. zsab146, 2021.

[56] M. Subramaniyan, F. G. Vital-Lopez, T. J. Doty, I. Anlap, W. D. S. Killgore, and J. Reifman, "Personalized alertness prediction using video-based ocular and facial features," *Sleep*, vol. 48, no. 11, p. zsaf149, 2025.

[57] C. B. S. Maior, M. J. d. C. Moura, J. M. M. Santana, and I. D. Lins, "Real-time classification for autonomous drowsiness detection using eye aspect ratio," *Expert Systems with Applications*, vol. 158, p. 113505, 2020.

[58] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research," in *Human Mental Workload*, ser. Advances in Psychology, P. A. Hancock and N. Meshkati, Eds. North-Holland, 1988, vol. 52, pp. 139–183.

[59] T. Åkerstedt, M. Ingre, G. Kecklund, A. Anund, D. Sandberg, M. Wahde, P. Philip, and P. Kronberg, "Reaction of sleepiness indicators to partial sleep deprivation, time of day and time on task in a driving simulator: The DROWSI project," *Journal of Sleep Research*, vol. 19, no. 2, pp. 298–309, 2010.

[60] A. E. Bryson and L. J. Henrikson, "Estimation using sampled data containing sequentially correlated noise," *Journal of Spacecraft and Rockets*, vol. 5, no. 6, pp. 662–665, 1968.

[61] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[62] J. Z. Bakdash and L. R. Marusich, "Repeated measures correlation," *Frontiers in Psychology*, vol. 8, p. 456, 2017.

[63] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986.

[64] C. Xiang, X. Fan, D. Bai, K. Lv, and X. Lei, "A resting-state EEG dataset for sleep deprivation," OpenNeuro dataset ds004902, version 1.0.4, 2024.

[65] H. P. A. Van Dongen, M. D. Baynard, G. Maislin, and D. F. Dinges, "Systematic interindividual differences in neurobehavioral impairment from sleep loss: Evidence of trait-like differential vulnerability," *Sleep*, vol. 27, no. 3, pp. 423–433, 2004.

[66] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950.

[67] F. Chung, B. Yegneswaran, P. Liao, S. A. Chung, S. Vairavanathan, S. Islam, A. Khajehdehi, and C. M. Shapiro, "STOP questionnaire: A tool to screen patients for obstructive sleep apnea," *Anesthesiology*, vol. 108, no. 5, pp. 812–821, 2008.

[68] M. W. Johns, "A new method for measuring daytime sleepiness: The Epworth sleepiness scale," *Sleep*, vol. 14, no. 6, pp. 540–545, 1991.

[69] C. H. Bastien, A. Vallières, and C. M. Morin, "Validation of the Insomnia Severity Index as an outcome measure for insomnia research," *Sleep Medicine*, vol. 2, no. 4, pp. 297–307, 2001.

[70] J. A. Horne and O. Östberg, "A self-assessment questionnaire to determine morningness-eveningness in human circadian rhythms," *International Journal of Chronobiology*, vol. 4, no. 2, pp. 97–110, 1976.

[71] J. Brooke, "SUS: A 'quick and dirty' usability scale," in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. London: Taylor & Francis, 1996, pp. 189–194.

[72] G. Piaggio, D. R. Elbourne, S. J. Pocock, S. J. W. Evans, and D. G. Altman, "Reporting of noninferiority and equivalence randomized trials: Extension of the CONSORT 2010 statement," *JAMA*, vol. 308, no. 24, pp. 2594–2604, 2012.

[73] R. J. Carroll, D. Ruppert, L. A. Stefanski, and C. M. Crainiceanu, *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd ed. Boca Raton, FL: Chapman & Hall/CRC, 2006.

[74] S. G. Hart, "NASA-Task Load Index (NASA-TLX); 20 years later," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 50, no. 9, pp. 904–908, 2006.

[75] F. Paas, J. E. Tuovinen, H. Tabbers, and P. W. M. Van Gerven, "Cognitive load measurement as a means to advance cognitive load theory," *Educational Psychologist*, vol. 38, no. 1, pp. 63–71, 2003.